

The First Break: Career Effects of Initial Access Shocks in Tournament Labor Markets*

Hugo Sant'Anna[†]

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Abstract

What is a randomly assigned opportunity worth in a labor market where access is rationed by rank? I study lucky losers in professional tennis, players who lose in final-round qualifying and enter the main draw when an accepted player withdraws, exploiting the lottery that Grand Slams use to fill such vacancies among the top four qualifying losers ($N = 120$ ATP, $N = 82$ WTA; a control-function design extends the analysis to $N = 2,054$ ATP and $N = 1,423$ WTA non-Grand-Slam events). Entry raises ranking points by 72 to 83 through 12 weeks on the ATP tour, surviving randomization inference, with a weaker 8-week gain of 55 points on the WTA tour and no effect on either tour within a year. Twelve pre-treatment placebo tests are indistinguishable from zero, and Lee bounds correcting for differential attrition stay positive through 26 weeks on the ATP tour. The gain operates through access rather than play: Elo, which adjusts for opponent quality, is flat on the ATP tour at every horizon, so the ranking gain reflects more draws entered rather than better play. Returns are stable across the first two opportunities and collapse at the third, so the decline in the return to opportunity is real but arrives later than a first-mover account implies. A Grand Slam slot pays about \$94,000 directly, while the downstream access cascade it triggers is worth roughly \$7,300. Persistence appears to depend on the market's record-keeping horizon: a rolling 52-week ranking window retires the initial shock on a known schedule, unlike the permanent records in settings where initial-condition effects run for a decade.

JEL Codes: J24, J44, L83

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[†]Department of Economics, Emory University. Email: hugo.santanna@emory.edu.

1 Introduction

In the summer of 2019, Brayden Schnur lost in the final round of Wimbledon qualifying and was, by the ordinary operation of the draw, finished for the fortnight. Borna Coric then withdrew from the main draw with a back injury, and Schnur was placed into the vacancy as a lucky loser. He played Wimbledon; the players he had been level with three days earlier did not. Professional tennis produces this natural experiment several times a year, and at Grand Slam events it produces it by literal lottery.

The question this makes answerable is what an opportunity is worth in a labor market where opportunity is rationed by rank. Tournament labor markets in the sense of [Lazear and Rosen \(1981\)](#) pay for relative rather than absolute performance, and the theoretical literature on them studies effort within a contest whose entrants are taken as given. Tennis runs a sequence of contests in which rank determines entry and entry determines the chance to earn rank, so a shock to one propagates into the other. A player granted a main-draw slot earns points, the points improve their ranking, and the improved ranking admits them to tournaments they would have had to qualify for. Whether that feedback loop converts a single administrative accident into a lasting career difference is an empirical question about how long the market's memory is.

I study lucky losers on the ATP and WTA tours: players who lose in the final round of qualifying and enter the main draw when an accepted player withdraws. At Grand Slams the vacancy is filled by a random draw among the top four ranked qualifying losers when the withdrawal precedes the end of qualifying, which supplies the randomization. I restrict the analysis to players who have never previously received a lucky loser slot, so treated and control players alike enter the pool with no prior exposure to this form of access. Two features of the setting support a decomposition that is rarely available. Ranking points reward participation without adjusting for the quality of the opponent; Elo ratings adjust for it. Tracking both after the same randomized shock separates a pure access effect from an improvement in play.

The empirical strategy has three components. Grand Slam lottery assignment provides the primary identification, with outcomes measured at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks and event fixed effects restricting comparisons to within an event’s top-four pool. A control-function approach extends the analysis to non-Grand-Slam events, where selection follows a ranking rule: the generalized residual is built from the exact enumeration probability that a player’s rank among qualifying losers falls within the available slots, computed over the full combinatorial structure of the qualifying draw. A match-level model then tests whether entry changes win probability conditional on opponent strength.

Lottery selection raises ranking points by 72.5 at 4 weeks, 72.1 at 8 weeks, and 82.8 at 12 weeks in the first-time ATP Grand Slam sample ($N = 120$), and each survives Fisher randomization inference at 5,000 permutations ($p_{\text{Fisher}} = 0.006, 0.022, 0.025$). The 26-week estimate of 72.6 clears the threshold with clustered standard errors ($p = 0.047$) but under neither the wild cluster bootstrap ($p_{\text{boot}} = 0.052$) nor Fisher inference ($p_{\text{Fisher}} = 0.103$), and the effect is indistinguishable from zero at 52 weeks. On the WTA tour ($N = 82$) the estimated gain concentrates at 8 weeks (+55.0), though it is significant only under clustered inference ($p = 0.017$) and not under randomization inference ($p_{\text{Fisher}} = 0.077$); Elo improves at medium horizons on that tour. Non-Grand-Slam control-function estimates on larger samples ($N = 2,054$ ATP; $N = 1,423$ WTA) reproduce the ATP pattern. Two checks carry most of the weight in supporting a causal reading: pre-treatment placebo regressions at -12 , -8 , and -4 weeks are indistinguishable from zero in all twelve tests, and Lee bounds correcting for differential attrition are tight and stay positive through 26 weeks on the ATP tour.

The evidence is not uniform in strength, and Section 7.7 sets out the ranking explicitly. The non-Grand-Slam access effects survive Romano-Wolf family-wise error control across the full 80-test outcome set and are the most secure results in the paper. The Grand Slam ranking-point effects survive randomization inference and false-discovery-rate correction but not family-wise control over that set, which is a demanding standard for a design with 120

observations and which treats a dynamic path across horizons as five independent claims rather than one. I report both standards throughout rather than the more favourable one.

This paper makes four contributions. First, it provides causal estimates of a within-career opportunity shock in a tournament labor market, with a decomposition into access and ability that the closest existing design cannot deliver. [Pallais \(2014\)](#) randomizes first job opportunities on an online labor market and finds persistent employment gains transmitted through a public rating, which is the same mechanism a tennis ranking implements. The advances here are that outcomes are observed at five horizons rather than one, so the trajectory is identified rather than assumed; that the market reports two distinct measures of a worker, which separates access from ability; that the effect is observed to decay, at a rate the market’s rolling window determines; and that some players receive the same randomized shock more than once, which identifies how the return varies with the number already received.

Second, the access-versus-ability decomposition finds that ranking gains on the ATP tour come from more draws entered and more matches played rather than from better play: Elo is flat at every horizon. The evidence points weakly toward over seeding, in which a player whose ranking runs ahead of their ability meets stronger opponents, but it remains suggestive: the match-level estimate is negative and imprecise ($\hat{\delta} = -0.141$, player-clustered $p = 0.17$), and the verified-lottery subsample shows an Elo decline only at 52 weeks (-51.3 , $p = 0.074$). The WTA tour behaves differently, with Elo rising at several horizons in the pooled sample; a direct test rules out the most obvious alternative explanation, since treated WTA players face stronger rather than weaker opponents after the focal event.

Third, the return to an opportunity declines with the number already received, but not where an emphasis on the first shock would predict. In a saturated specification estimated on the pooled sample, the second-lucky-loser shifter is statistically indistinguishable from zero at every horizon, while the third-plus shifter is large, negative, and significant throughout; a joint test rejects a flat experience profile on three of four samples. Returns are stable across

the first two opportunities and collapse at the third, consistent with a direct-entry threshold that a single main-draw appearance is often insufficient to clear.

Fourth, the results separate two economic magnitudes that are easily conflated. A Grand Slam main-draw slot pays about \$94,000 more than a qualifying exit, which is one to two years of a qualifying-level player’s typical earnings; this is what the tournament pays for the slot and is mechanical given the round reached. The causal cascade, the additional main draws entered over the following 26 weeks, is worth roughly \$7,300, about 8 percent of that immediate figure. On non-Grand-Slam events, where the immediate prize is far smaller, the ordering reverses.

The access gain fades within a year, which contrasts with the decade-long persistence documented for initial labor market conditions (Oyer, 2006; Oreopoulos et al., 2012). Section 8 argues that the difference lies in the market’s record-keeping horizon rather than in the size of the shock: the 52-week rolling ranking window mechanically retires the points that created the advantage, whereas a permanent record has no comparable expiry.

Section 2 reviews the related literature. Section 3 describes the institutional setting and data. Section 4 presents the econometric framework. Section 5 reports the main results. Section 6 examines mechanisms. Section 7 presents robustness checks, including the experience gradient and the verified-lottery subsample. Section 8 concludes.

2 Related Literature

2.1 Tournament Labor Markets

The setting studied here is a tournament labor market in the sense of Lazear and Rosen (1981): compensation and advancement depend on rank rather than on absolute output. Rank-order tournaments economize on measurement when absolute performance is costly to observe, and Rosen (1986) shows that in elimination structures the prize spread must widen at later rounds to maintain effort, because the option value of continuing shrinks as the

field narrows. [Prendergast \(1999\)](#) surveys the resulting incentive literature. Empirical tests have relied heavily on sport, where rank, prize, and effort are all measured: [Ehrenberg and Bognanno \(1990\)](#) show that prize spread affects scoring in professional golf, [Sunde \(2009\)](#) finds that heterogeneity between competitors dampens effort in tennis, and [Genakos and Pagliero \(2012\)](#) show that interim rank shapes risk-taking in dynamic tournaments. [Kahn \(2000\)](#) sets out the general case for sport as a labor-market laboratory.

This paper takes the tournament structure as given and asks a question the tournament literature has not addressed: what happens when *access* to the tournament is itself randomly assigned. Standard tournament models treat the set of competitors as fixed and study effort within a contest. Professional tennis instead runs a sequence of contests in which yesterday’s rank determines today’s entry, so a shock to rank propagates as a shock to future participation. The object of interest is therefore the return to an entry slot rather than the elasticity of effort with respect to the prize spread.

2.2 Randomized Access Shocks

The closest analog is [Pallais \(2014\)](#), who randomly assigns first job opportunities to inexperienced workers on an online labor market and finds large, persistent employment gains. The mechanism she identifies is informational: the randomly granted job generates a public rating that subsequent employers use, so the market’s own record-keeping converts a one-time assignment into a durable advantage. The tennis ranking plays the same role as the oDesk rating, and the present setting shares the essential design feature of literal randomization.

Four features of this setting let the analysis go beyond what the oDesk design supports, and they define the paper’s contribution relative to [Pallais \(2014\)](#). First, outcomes are observed at five horizons from 4 to 52 weeks rather than at a single follow-up, so the trajectory of the effect is identified rather than assumed. Second, the market reports two distinct measures of a worker: ranking points, which reward participation without adjusting for opponent quality, and Elo, which adjusts for it. Their divergence separates an access channel

from an ability channel, a decomposition that a single reputational score cannot deliver. Third, the effect is observed to decay, and the decay rate is informative about the market’s memory: the 52-week rolling ranking window mechanically retires the initial shock, whereas an oDesk rating persists indefinitely. Fourth, some workers receive the same randomized shock more than once, which identifies how the return to an opportunity varies with the number already received.

Related evidence on access shocks comes from [Abel et al. \(2020\)](#), who randomize reference letters for South African job seekers, and [Startz \(2021\)](#), who exploits random interviewer assignment. In science, [Azoulay et al. \(2014\)](#) document cumulative advantage following early recognition, [Wang et al. \(2019\)](#) find that narrowly missing early funding predicts stronger later performance, and [Azoulay et al. \(2010\)](#) show that losing access to a star collaborator reduces output. Threshold-crossing designs in education find long-run earnings gains from marginal admission ([Zimmerman, 2014](#); [Hoekstra, 2009](#); [Goodman et al., 2017](#)).

2.3 Initial Conditions and Path Dependence

A large literature establishes that early-career conditions have durable effects: graduating in a recession depresses earnings for a decade ([Oreopoulos et al., 2012](#)), initial placement shapes the subsequent research careers of economists ([Oyer, 2006](#)), MBA cohort timing determines entry into investment banking ([Oyer, 2008](#)), and childhood location shapes intergenerational mobility ([Chetty et al., 2014](#)). [Malmendier and Nagel \(2011\)](#) show that early macroeconomic experience alters risk preferences decades later, which is the sharpest existing statement of the claim that a first experience is qualitatively different from a later one. The common mechanism is cumulative advantage ([Merton, 1968](#); [Rosen, 1981](#)).

This paper differs from that literature in the nature of the shock and in the market that transmits it. The shocks studied there are shocks to the environment a career starts in; the shock here occurs within an ongoing career and is randomly assigned by an institutional rule. The persistence results also differ, and the difference is informative: where recession-cohort

effects run for a decade, the access gains documented here dissipate within a year. Section 8 argues that this contrast identifies market transparency, and specifically the presence of a rolling performance window, as the moderator governing how long an access shock survives.

2.4 Sports as an Empirical Laboratory

Gauriot and Page (2019) use narrowly decided tennis matches to show that ranking systems reward luck as though it were skill, which motivates the separation of ranking points from Elo pursued here. González-Díaz et al. (2012) study performance under pressure and Berger and Pope (2011) examine responses to interim deficits in basketball.

Most directly, Maity et al. (2025) study lucky losers across several professional sports and report that early setbacks correlate with stronger subsequent performance. Their comparison is between lucky losers and other qualifying losers without a design that accounts for how lucky losers are selected, so the estimates combine the effect of entry with the selection that produced it. The Grand Slam lottery provides the missing randomization, and Section 6.4 shows that the correlation they document is consistent with an access channel rather than a performance channel.

3 Institutional Setting and Data

3.1 Institutional Background

Professional tennis is organized into a hierarchical tournament structure by the Association of Tennis Professionals (ATP, men) and the Women’s Tennis Association (WTA, women). Four Grand Slam tournaments (the Australian Open, Roland Garros, Wimbledon, and the US Open) sit at the top of the hierarchy, awarding up to 2,000 ranking points for the winner. Below them are Masters 1000 events, followed by ATP/WTA 500 and 250 events. Each tournament awards ranking points based on the round reached, with higher-tier events offering more points per round. Both tours use a rolling 52-week ranking: a player’s ranking

equals the sum of points from their best results over the previous year, updated weekly. Rankings determine direct entry into tournament main draws: the top 104 players enter Grand Slam main draws directly, the top 40–56 enter Masters events, and similar thresholds apply at lower tiers. This structure creates a feedback loop central to the analysis: ranking points determine tournament access, and tournament access determines opportunities to earn ranking points.

Players ranked below the direct-entry threshold can enter a tournament’s qualifying draw, a mini-tournament held before the main event. Qualifying draws are single-elimination brackets (typically 32 or 16 players on the ATP tour, 12 or 16 on the WTA tour), with 4 to 8 qualifiers advancing to the main draw after winning two or three consecutive matches. The players who populate qualifying draws are ranked approximately 100th to 300th in the world, too low for direct entry but competitive enough to warrant a spot in the qualifying field. These marginal players are the focus of this study. The two tours differ in ways that matter for the analysis. The WTA calendar is denser, with more tournaments per month and shorter gaps between events. WTA qualifying draws are typically smaller (often 12 players compared to 16 on the ATP side), so fewer final-round qualifying losers are generated per event.

When a player accepted into a tournament main draw withdraws (due to injury, illness, or scheduling conflicts), the vacant slot is filled by a lucky loser: a player who lost in the final round of qualifying. The selection mechanism differs by tournament tier, and this difference is central to the identification strategy. At Grand Slam tournaments, the rulebook specifies a two-track system based on withdrawal timing. When a main-draw withdrawal occurs *before* qualifying is complete, the top four ranked final-round qualifying losers enter a random draw (lottery) for the available slot; each of the four has an equal probability of selection regardless of ranking differences within the group. When a withdrawal occurs *after* qualifying ends, the highest-ranked final-round qualifying loser receives the slot without a lottery. The lottery pathway commenced after Wimbledon 2005 and provides the basis for

the primary identification strategy. At non-Grand-Slam events (Masters 1000, ATP 500, ATP 250), lucky loser selection follows a strict ranking rule: the highest-ranked final-round qualifying loser receives the first available slot, the second-highest receives the next, and so on. There is no lottery component.

I supplement ranking points with Elo ratings as a measure of playing ability independent of the ranking-point system. Elo ratings update after each match based on the outcome and both players’ pre-match ratings; beating a higher-rated opponent produces a larger gain than beating a lower-rated one. I compute Elo ratings from the full match history using a K -factor of 32 and an initial rating of 1,500. The distinction between ranking points (which reward participation regardless of opponent quality) and Elo ratings (which adjust for opponent quality) is central to the mechanism analysis: if LL entry improves rankings but not Elo, the improvement reflects more participation rather than improved ability.

3.2 Data Sources and Sample Construction

The primary data source is Jeff Sackmann’s Tennis Abstract database, a comprehensive public dataset of professional tennis match results covering the ATP tour from 1968 to 2024 and the WTA tour over the same period. The database includes match outcomes, player rankings, tournament details (surface, tier, draw size), and entry type (direct acceptance, qualifying, wildcard, or lucky loser). I supplement match records with weekly ranking files to construct ranking trajectories before and after each event. Elo ratings are computed from the full match history using the parameters described above.

The unit of observation is a player-episode pair (i, e) , where player i is a final-round qualifying loser at event e . The eligible pool consists of all final-round qualifying losers at Grand Slam events where at least one lucky loser slot was awarded, restricted to the top four ranked losers per event and to players with *no prior lucky loser awards of any type*. This LL-naive restriction ensures that both treated and control players enter the lottery pool without any prior main-draw access through the LL mechanism. Players at events with zero

LL entries are excluded because the absence of main-draw withdrawals means no player at that event faced any probability of receiving LL treatment. Grand Slam qualifying data with lottery-based assignment are available from 2006 to 2024.

The treatment variable is defined as

$$D_{ie} = 1 \text{ if player } i \text{ entered the main draw as a lucky loser at event } e.$$

For the Grand Slam lottery design, treatment assignment is random within the top-four pool. For the non-Grand-Slam control-function design, treatment is determined by the ranking rule, and endogeneity is corrected using a control function derived from the peer component of the selection probability.

The primary estimation sample consists of first-time LL candidates among the top-four ranked final-round qualifying losers at Grand Slams from 2006 to 2024. Both treated and control players have zero prior LL awards of any type (Grand Slam or non-Grand-Slam), so the estimated effect captures the impact of a first-ever access shock. The first-time ATP Grand Slam sample contains $N = 120$ player-episode observations (61 treated, 59 control) across 41 events with identifying variation. The first-time WTA Grand Slam sample contains $N = 82$ observations (33 treated, 49 control). The *first-time non-Grand-Slam sample* applies the same LL-naive restriction and contains $N = 2,054$ ATP observations (523 treated, 1,531 control) across 266 events with variation, and $N = 1,423$ WTA observations (378 treated, 1,045 control) across 258 events.¹ The non-Grand-Slam control function is re-estimated on the first-LL-restricted event set. The robustness section reports a marginal impact analysis that pools all LL recipients (including repeat ones) and compares treatment effects across experience levels (Section 7.1).

¹WTA 1000 events (formerly Premier Mandatory) correspond to ATP Masters 1000; WTA 500 (formerly Premier) to ATP 500; WTA 250 (formerly International) to ATP 250.

3.3 Lucky Loser Distribution

LL entries are not rare events in the first-LL sample. Across the 41 ATP Grand Slam events with identifying variation, 61 of 120 top-four LL-naive pool members received LL entry, a rate of 51 percent. On the WTA side, 33 of 82 pool members (40 percent) received entry across 32 events. The lower WTA rate reflects smaller qualifying draws and fewer withdrawal opportunities. In the non-Grand-Slam sample, 523 of 2,054 ATP pool members received LL entry (25 percent) and 378 of 1,423 WTA members received entry (27 percent); the lower rates reflect larger pool sizes at ATP 250/500 and WTA 250/500 events.

3.4 Variable Construction

Outcomes are organized into two categories. *Immediate event-level outcomes* measure the direct effect of the LL slot at the focal event: whether the player won at least one main-draw match, the number of main-draw matches played, and ranking points earned at the event. *Post-episode dynamic outcomes* measure career trajectories at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks following the event. Primary dynamic outcomes include the number of main-draw entries (MainDraws_{ie}^h), the number of matches played at ATP/WTA 250+ events (Matches250_{ie}^h), and the change in ranking points ($\Delta\text{Points}_{ie}^h$). Secondary outcomes include the change in Elo rating (ΔElo_{ie}^h). The Elo outcome is the key test of ability: if LL entry improved playing skill, Elo ratings would rise.

Pre-draw state variables enter as controls:

$$\begin{aligned} Z_{ie}^{pre} = \{ & \text{Points}_{ie}/1000, (\text{Points}_{ie}/1000)^2, \\ & \text{Elo}_{ie}/100, (\text{Elo}_{ie}/100)^2, \\ & \text{SurfElo}_{ie}/100, (\text{SurfElo}_{ie}/100)^2, \text{Age}_i \} \end{aligned}$$

Pre-treatment controls include ranking points / 1000 (linear and quadratic), Elo rating / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and player age. Prior-

LL count variables are excluded because they are zero by construction for all players in the first-LL-restricted sample. Elo ratings are divided by 100 when used as covariates to improve coefficient scale (raw Elo ranges from about 1,400 to 2,200); Elo *outcomes* (change in Elo rating) remain in raw points for direct interpretability. Ranking points and Elo absorb the mechanical confound that higher-ranked LL candidates have better tournament access regardless of LL entry. Age captures career-stage effects. In pooled ATP–WTA specifications, a tour indicator is added.

Only pre-treatment variables are included as controls. Conditioning on post-treatment variables (for example, contemporaneous Elo or ranking at horizon h) would absorb part of the causal pathway. Given the sample sizes (120 observations in the first-LL ATP lottery sample, 82 in the WTA sample), I favor simpler models with fewer explanatory variables to preserve degrees of freedom.

3.5 Summary Statistics

Summary statistics for the ATP and WTA first-LL Grand Slam estimation samples are reported in Appendix Tables 17 and 18.

Players in the Grand Slam lottery pool are ranked approximately 100th to 250th in the world, with a mean Elo rating around 1,700. Ages range from 18 to 35, with a mean of approximately 24 years. The distribution is right-skewed in ranking, with a few players near the top 100 mixed with many ranked 150–250, and approximately symmetric in Elo.

The balance tests (reported in Section 7.4) show that the first-time ATP sample has a joint F -test p -value of 0.106 and the first-time WTA sample has a joint F -test p -value of 0.063. Within-event balance tests yield $p = 0.046$ (ATP) and $p = 0.516$ (WTA); the within-event test is the economically relevant specification since randomization operates within each event’s top-four pool.

The players in this sample are drawn from the same competitive margin: final-round qualifying losers ranked in the top four at their event, with no prior LL awards of any

type. They are heterogeneous in age, ranking, and career stage, and the randomization at Grand Slams ensures that treatment and control groups come from the same distribution. The first-time non-Grand-Slam samples have similar characteristics but are larger (ATP: $N = 2,054$; WTA: $N = 1,423$), giving greater statistical power at the cost of relying on the control function for identification. Non-Grand-Slam summary statistics are reported in the appendix.

4 Econometric Framework and Identification

4.1 Grand Slam Identification Strategy

I restrict both the Grand Slam and non-Grand-Slam samples to *first-time* lucky loser candidates: players who have never previously received a lucky loser slot of any type (Grand Slam or non-Grand-Slam). Both treated players ($D_{ie} = 1$, received the LL slot) and controls ($D_{ie} = 0$, eligible but not selected) are LL-naive: they have zero prior LL awards. This restriction ensures that the estimated effect captures the impact of a first-ever access shock, free of contamination from prior LL experience, and simplifies the pre-treatment control vector Z_{ie}^{pre} by eliminating prior-LL count variables that are zero by construction.

The Grand Slam lottery provides random assignment of lucky loser entry within the top-four pool. The identifying assumption is

$$E[Y_{ie}(0) \mid D_{ie} = 1, e] = E[Y_{ie}(0) \mid D_{ie} = 0, e],$$

so among the top four ranked final-round qualifying losers at a given Grand Slam, lottery-selected and unselected players have the same expected untreated potential outcomes. This holds by design: within the pool, each player has an equal probability of selection regardless of ranking, ability, or other characteristics.

The randomization operates within each event’s top-four pool. A player ranked 120th at

the Australian Open and a player ranked 180th at Roland Garros belong to different pools, so the lottery alone does not make them comparable. The event fixed effect α_e absorbs all between-pool variation, so the estimated effect compares lottery winners to lottery losers within the same event.

The lottery operates only when withdrawals occur before qualifying completes. Post-qualifying withdrawals follow the ranking rule, so the sample may include some ranking-based assignments. Section 7.5 tests this directly with pre-treatment placebo regressions, which is the sharpest available check because ranking-rule assignment selects on pre-treatment rank. Section 7.3 additionally isolates events where the assignment was provably a lottery, using a criterion that requires no external information: an entrant ranked below the number of available slots cannot have been placed by the ranking rule.

With only four players per pool and a binary treatment, finite-sample imbalance is expected even under true randomization. In the first-time ATP Grand Slam sample ($N = 120$), the joint F -test yields $p = 0.106$; on the WTA side ($N = 82$), $p = 0.063$. Neither rejects at the 5% level, though both are closer to marginal than in the full sample, reflecting the smaller first-LL-restricted pool. To improve precision and absorb residual imbalance, I include pre-treatment controls in all dynamic specifications; I also report specifications without controls as a complement.

4.2 Immediate Effects

The event-level model estimates the immediate effect of receiving the LL slot at the focal Grand Slam:

$$Y_{ie}^{imm} = \alpha_e + \delta D_{ie} + \varepsilon_{ie}, \quad (1)$$

where Y_{ie}^{imm} is one of three event-level outcomes: an indicator for winning at least one main-draw match, the number of main-draw matches played, and ranking points earned at the event. The event fixed effect α_e is a Grand Slam $\times$ Year indicator. The coefficient δ

measures the size of the initial treatment shock: the immediate dose of ranking points and competitive experience that the LL slot provides. This quantity sets the scale for interpreting the dynamic effects that follow. Control outcomes are mechanically zero because unselected players do not enter the main draw.

4.3 Post-Episode Dynamics

The dynamic model traces career outcomes at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks:

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h}, \quad (2)$$

where $Y_{ie,h}$ is the outcome for player i from event e measured at horizon h , α_e is an event fixed effect (Grand Slam $\times$ Year), α_h is a horizon fixed effect absorbing level differences across time horizons, Z_{ie}^{pre} is the vector of pre-treatment controls (ranking points / 1000 linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, and player age) with horizon-invariant coefficients Γ , and $\varepsilon_{ie,h}$ is the error term. Standard errors are clustered at the player level because players may receive repeated LL opportunities across events, inducing within-player dependence across episodes and horizons; event-level clustering is unnecessary because event fixed effects already absorb between-event variance. The coefficients on Z^{pre} do not vary by horizon because these controls absorb pre-treatment differences that are constant across the post-event trajectory. The horizon fixed effects are important for count outcomes (main draws entered, matches played) where level differences across horizons are substantial; adding α_h raises the R^2 from 0.35 to 0.53 for main-draw entries.

The coefficient β_h identifies the causal effect of lottery selection on the outcome at horizon h . The sequence $\{\beta_4, \beta_8, \beta_{12}, \beta_{26}, \beta_{52}\}$ traces the divergence between treated and control trajectories after the random shock. Immediate effects ($h = 0$) are captured by the event-level model in equation (1) and are excluded from the dynamic specification to avoid double-

counting.

The estimand β_h is the average treatment effect on the treated (ATT) at horizon h , averaged across events with weights proportional to the number of first-time LL candidates in the top-four pool. The event fixed effects ensure that comparisons are within-pool, and the stacked specification pools across horizons to improve precision. Because the lottery assigns treatment with equal probability within each pool, the ATT and ATE coincide within the experimental population.

A player may appear in the lottery pool at multiple events. To avoid contamination from subsequent treatments, I censor each player-episode at the date of the player’s next final-qualifying-round loss at an LL-granting event of the same type:

$$h^* = \min(12 \text{ months, time to next same-type qualifying loss at an LL-granting event}).$$

For Grand Slam episodes, this censors at the next Grand Slam qualifying loss where at least one LL slot was awarded; for non-Grand-Slam episodes, censoring occurs at the next non-Grand-Slam qualifying loss at an LL-granting event. Within this window, the only relevant treatment is D_{ie} from the originating event, so overlapping treatment spells do not contaminate the estimate. A player who appears in the LL candidate pool at multiple events generates multiple episodes, potentially with different treatment statuses. As a robustness check, I restrict the sample to each player’s first Grand Slam LL opportunity (Section 7.1).

The dynamic model’s 4-week horizon measures outcomes *after* the focal event. Because the focal event itself occupies the first 1–2 weeks of the post-treatment window, treated players may enter fewer *additional* main draws during this period than controls who are free to enter other events; the immediate-effects model (equation 1) captures the focal event’s outcomes separately. The negative 4-week main-draws coefficient in the non-GS sample reflects this mechanical timing effect; by 12 weeks the treatment effect on main draws becomes positive as ranking gains open new entry opportunities.²

²This timing pattern is specific to count outcomes (main draws entered, matches played) and does not

4.4 Heterogeneous and Treatment-Dose Effects

I estimate heterogeneous treatment effects by augmenting equation (2) with interactions between the treatment indicator D_{ie} and category dummies. For a categorical moderator with K groups (one base category omitted), the estimating equation is

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \sum_{k=2}^K \delta_{k,h} D_{ie} \times \mathbf{1}[G_{ie} = k] + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h}, \quad (3)$$

where β_h captures the base-category treatment effect at horizon h and $\delta_{k,h}$ captures the differential effect for category k relative to the base. Because β_h is already indexed by horizon in equation (2), this specification inherits the horizon-specific treatment structure without additional horizon interactions. I examine three dimensions: pre-treatment ranking (above versus below the tour-specific median, with below-median as base), age (below versus above 24, with younger as base), and prior LL experience (none versus any, with no prior experience as base).

To characterize how treatment effects vary with dose, I interact D_{ie} with the number of main-draw matches won at the focal event, entered continuously. The dose variable is endogenous: players who win more matches are better or luckier on the day, so these estimates are descriptive rather than causal. They answer a different question than the ITT: conditional on receiving the LL slot, how much does the career impact depend on what the player does with it? Standard errors are clustered at the player level throughout.

4.5 Non-Grand-Slam Control-Function Strategy

At non-Grand-Slam events, LL slots go to the highest-ranked final-round qualifying losers. I use a control-function approach to account for endogenous LL assignment, exploiting the peer component of the selection probability computed from the combinatorial structure of qualifying match pairings.

affect ranking-point or Elo outcomes, which cumulate over the horizon window inclusive of the focal event.

The selection probability P_i^{LL} is the probability that player i 's rank among final-round qualifying losers falls within the top c slots (where c is the number of LL positions available), conditional on player i having lost:

$$P_i^{LL} = \Pr(\text{rank among losers} \leq c \mid i \text{ loses}).$$

This probability depends on the qualifying match outcomes of *other* players, specifically on the probability that enough higher-ranked qualifiers also lose their matches. Construction proceeds in two steps. First, I estimate a weighted logit model of match outcomes on qualifying-round matches at non-Grand-Slam LL-granting events, separately for each tour (ATP: $N = 9,775$; WTA: $N = 10,199$ qualifying matches). The match-level covariates X_{ijm} include ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed head-to-head win proportion, head-to-head encounter count, and age difference, along with event-level surface indicators (clay and grass). The logit is estimated by weighted MLE, with weights equal to the total points in play (sum of service points) in each match; qualifying round number serves as a fallback weight when service point data are unavailable. Estimating on all qualifying rounds rather than the final round alone raises the effective sample for the win model from roughly 2,400 to 9,775 matches on the ATP side and 10,199 on the WTA side. The same logit specification is used for the selection probability, the tournament performance model, and the treatment-dose analysis, so the covariate structure is unified across applications. This yields predicted match-winning probabilities $\hat{P}(a_m \rightarrow b_m \mid X, Z)$ for each final-round qualifying match m between players a_m and b_m . Second, given these predicted probabilities and the $N_e - 1$ other final-round qualifying matches (excluding player i 's match, since i is conditioned on having lost), I enumerate all 2^{N_e-1} possible outcome combinations. Each combination has probability equal to the product of the individual match outcome probabilities from the logit, invoking conditional independence of match

outcomes given observables. Conditional independence is plausible because each final-round qualifying match involves a distinct pair of players with independent on-day shocks, and the logit already conditions on ranking, Elo, surface Elo, head-to-head history, age, and surface; residual correlation across same-event matches would have to arise from unobserved event-day conditions shared across disjoint player pairs, which is a priori weak. For each combination, the set of losers is determined and their rankings compared; P_i^{LL} is the sum of probabilities of all combinations in which i 's rank falls within the top c among losers. Player i 's own loss probability does not enter P_i^{LL} ; the enumeration uses only final-round matches, since the LL selection mechanism conditions on the final qualifying-round structure. The resulting P_i^{LL} is systematically lower than a naive Bernoulli convolution that ignores the draw structure (mean 0.279 vs. 0.506 across the non-GS sample), because the exact enumeration correctly conditions on which players are paired against each other. The key identifying assumption is that, conditional on player i 's characteristics and the fact that i lost, the qualifying-match outcomes of other players affect i 's future career only through the lucky-loser channel.

For all non-Grand-Slam specifications, I account for endogenous LL assignment using a control-function approach. The generalized residual $\hat{v}_{ie}$ is computed analytically from P_i^{LL} :

$$\hat{v}_{ie} = D_{ie} \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{P_i^{LL}} - (1 - D_{ie}) \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{1 - P_i^{LL}},$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal PDF and CDF. This residual captures how far player i 's actual LL status deviates from what P_i^{LL} would predict under normality. It enters the outcome equation as an additional regressor:

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \rho \hat{v}_{ie} + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h}. \quad (4)$$

The coefficient $\hat{\rho}$ tests for endogeneity of LL entry: $\hat{\rho} \neq 0$ indicates that LL assignment is correlated with unobservables and the control-function correction is needed. The ATP

non-Grand-Slam sample contains $N = 2,054$ observations (523 treated); the WTA sample contains $N = 1,423$ (378 treated). Standard errors are clustered at the player level.

For the match-level tournament performance model at non-Grand-Slam events (Section 6.2), the same analytically computed generalized residual enters the outcome logit:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta \cdot D_{ie} + \rho \cdot \hat{v}_{ie}), \quad (5)$$

where X_{ijm} includes ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed H2H win proportion, H2H encounter count, age difference, and surface indicators (clay and grass): the same logit specification used for the selection probability. Z_{ie}^{pre} includes ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and player age. In the dose specification, $\hat{v}_{ie}$ is interacted with the centered matches-won variable to allow the endogeneity correction to vary with treatment dose, because players who win more matches at the LL event may have different unobserved ability profiles. Standard errors are clustered at the player level; the same clustering applies to the delta specification and preserves within-player dependence across events and matches.

5 Results

Grand Slam results exploit the lottery for identification and are presented separately for the ATP and WTA tours. Non-Grand-Slam control-function results are summarized in Section 5.4 and reported in full in the appendix.

5.1 Immediate Effects

Table 1 characterizes the treatment dose at the focal event. Because control players do not enter the main draw, these estimates describe the magnitude of the treatment rather than a causal effect against a counterfactual.

On the ATP tour ($N = 120$), treated players play additional main-draw matches and earn ranking points proportional to the round reached; the WTA pattern ($N = 82$) is similar. A Grand Slam first-round loss earns 10 ranking points and a second-round loss earns 45 points, so even a single main-draw win delivers an economically meaningful shock to the player’s rolling 52-week ranking-point stock.

Table 1: Grand Slam Lottery: Immediate Event-Level Effects

Outcome	<i>t</i> -test	Event FE	
	LL Mean	$\hat{\beta}$	(SE)
<i>ATP</i>			
Main-draw match win	0.34***	0.39***	(0.09)
Main-draw matches played	1.46***	1.64***	(0.13)
Ranking points at event	28.85***	36.35***	(5.50)
<i>N</i>	61 / 59	120	
<i>WTA</i>			
Main-draw match win	0.36***	0.33***	(0.09)
Main-draw matches played	1.42***	1.38***	(0.12)
Ranking points at event	25.45***	23.93***	(4.35)
<i>N</i>	33 / 49	82	
<i>Pooled</i>			
Main-draw match win	0.35***	0.38***	(0.06)
Main-draw matches played	1.45***	1.53***	(0.09)
Ranking points at event	27.66***	31.32***	(3.97)
<i>N</i>	94 / 108	202	

Notes. First-time Grand Slam LL candidates: top-four ranked final-round qualifying losers at events with at least one LL slot, 2006–2024. ATP: $N = 120$; WTA: $N = 82$. Event FE columns report OLS with Grand Slam $\times$ Year fixed effects. Control outcomes are mechanically zero because unselected players do not enter the main draw. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Post-Episode Dynamics

Tables 2 and 3 report estimates from a stacked specification with horizon-specific treatment effects estimated jointly. The stacked model pools all five horizons into a single regression with horizon-specific LL coefficients and horizon fixed effects, improving efficiency by borrowing information across horizons while allowing the treatment effect to vary freely at each

horizon.

On the ATP side, the sample comprises 120 first-time player-episodes drawn from 109 distinct players across 55 events, contributing approximately 600 stacked observations over five horizons. The lottery raises ranking points by 72.5 at 4 weeks, 72.1 at 8 weeks, and 82.8 at 12 weeks, before peaking and starting to fade: the effect is 72.6 at 26 weeks and becomes indistinguishable from zero at 52 weeks. Fisher randomization inference, computed from 5,000 permutations within Grand Slam $\times$ Year blocks, supports the first three horizons ($p_{\text{Fisher}} = 0.006, 0.022, 0.025$), so those results do not depend on the asymptotic approximation. The permutation retains only blocks containing both a treated and a control observation, since singleton blocks contribute no permutation variation, which reduces the effective sample from 120 to 103 on the ATP tour and from 82 to 77 on the WTA tour. The 26-week estimate clears the threshold with clustered standard errors ($p = 0.047$) but under neither the wild cluster bootstrap ($p_{\text{boot}} = 0.052$) nor Fisher inference ($p_{\text{Fisher}} = 0.103$), which makes it the least robust of the significant effects and the one that should carry least weight. Benjamini-Hochberg q -values across the 80-test family of primary outcomes appear in Appendix Table 32, and Romano-Wolf step-down p -values, which account for the dependence across horizons that BH ignores, appear alongside them in Table 34. Elo rating changes are null at every horizon, so the ranking-point gains come from more matches and rounds played rather than from measured ability improvement. The trajectory is consistent with a real but temporary point shock: it accumulates through 12 to 26 weeks as the ranking boost opens further tournaments, then dissipates as the original points age out of the rolling window.

On the WTA side, the 82-episode sample delivers approximately 410 stacked observations. Ranking points rise by 55.0 at 8 weeks, which is significant under clustered inference ($p = 0.017$) but not under randomization inference ($p_{\text{Fisher}} = 0.077$); estimates at other horizons are positive but imprecise. Given that Fisher inference is the more appropriate standard for a lottery of this size, the WTA ranking-point result should be read as suggestive rather than established. The Elo pattern differs from the ATP tour: WTA first-timers show Elo

gains at 8 weeks (+15.0, $p = 0.082$) and at 52 weeks (+36.8, $p = 0.055$). On the verified-lottery subsample the point estimates keep their sign and rough magnitude but none is significant, which the minimum detectable effects for a 41-observation subsample lead one to expect (Section 7.3). These estimates should be read against the borderline joint balance ($p = 0.063$) and the smaller sample. The Elo channel on the WTA is examined in Section 6.

Table 2: ATP Grand Slam: Post-Episode Dynamic Effects (Stacked Model)

	4w	8w	12w	26w	52w
Ranking Pts Δ	72.5*** (26.7)	72.1*** (26.6)	82.8*** (26.1)	72.6** (36.2)	23.5 (59.1)
Elo Δ	-4.4 (9.1)	0.2 (10.9)	-4.8 (10.3)	4.7 (13.5)	-9.7 (21.0)
Main Draws	-0.2 (0.5)	0.1 (0.5)	0.1 (0.5)	0.7 (0.5)	0.8 (1.5)
Matches 250+	0.3 (1.0)	0.7 (1.0)	0.9 (0.9)	1.5 (1.1)	1.3 (2.8)
Win %	0.04 (0.05)	-0.01 (0.03)	-0.03 (0.03)	-0.03 (0.03)	-0.03 (0.03)

Notes. Stacked specification: 120 first-time player-episodes contributing approximately 600 stacked observations across five horizons (4, 8, 12, 26, 52 weeks). Horizon-specific LL treatment effects estimated jointly with horizon fixed effects, Grand Slam $\times$ Year fixed effects, and pre-treatment controls (ranking points linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, and age). Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figures 1 and 2 plot mean cumulative ranking points earned by LL-selected and unselected players from 12 weeks before to 52 weeks after each Grand Slam. Pre-event trajectories on both tours are parallel, so pre-treatment divergence is not detectable. After the event, ATP LL recipients accumulate approximately 72 additional ranking points by 4 weeks and 83 by 12 weeks; the advantage then narrows as the rolling window absorbs the initial gains. WTA LL recipients diverge at 8 weeks by 55 points, with controls catching up more quickly than on the ATP tour, a pattern consistent with the denser WTA calendar.

Table 3: WTA Grand Slam: Post-Episode Dynamic Effects (Stacked Model)

	4w	8w	12w	26w	52w
Ranking Pts Δ	40.0 (28.5)	55.0** (22.5)	27.8 (20.1)	7.5 (42.8)	26.5 (86.4)
Elo Δ	13.3* (7.6)	15.0* (8.5)	11.8 (9.1)	21.6* (12.7)	36.8* (18.9)
Main Draws	-0.1 (0.5)	0.3 (0.5)	0.3 (0.6)	0.7 (0.7)	1.8 (1.5)
Matches 250+	-0.5 (1.0)	0.8 (0.7)	0.6 (0.8)	1.9* (1.0)	3.2 (2.5)
Win %	0.03 (0.05)	0.05* (0.03)	0.05* (0.03)	0.04 (0.03)	0.05 (0.03)

Notes. Stacked specification: 82 first-time player-episodes contributing approximately 410 stacked observations across five horizons (4, 8, 12, 26, 52 weeks). Horizon-specific LL treatment effects estimated jointly with horizon fixed effects, Grand Slam $\times$ Year fixed effects, and pre-treatment controls. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

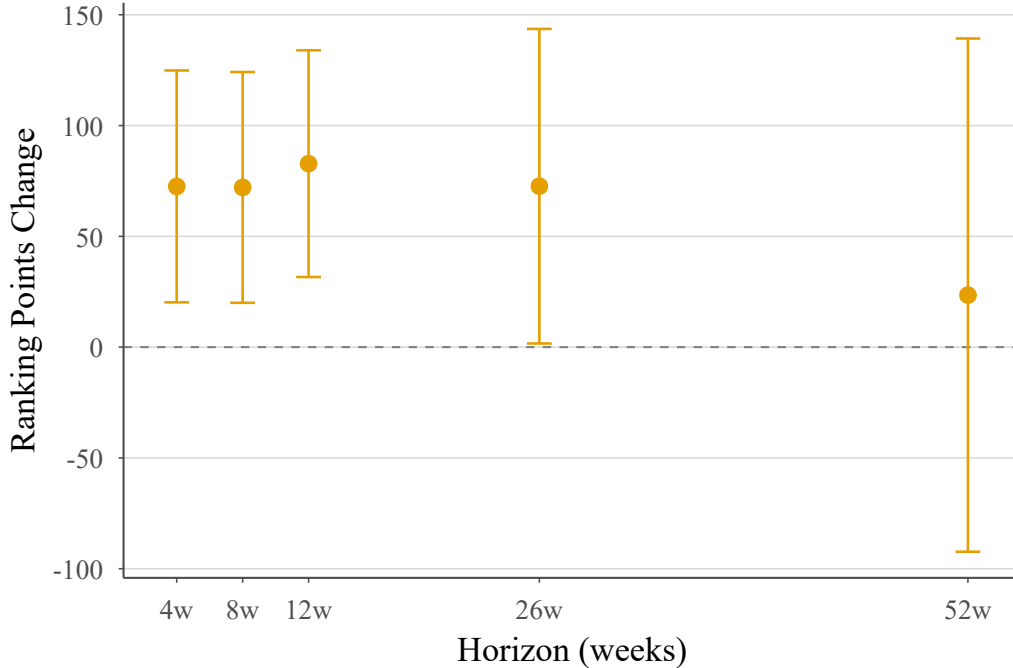


Figure 1: ATP Event Study: Ranking Trajectories for Selected and Unselected Players

Notes. Each point is the estimated treatment effect at the indicated horizon. Vertical dashed line marks the event date. Shaded regions are 95 percent confidence intervals. First-time ATP Grand Slam sample ($N = 120$).

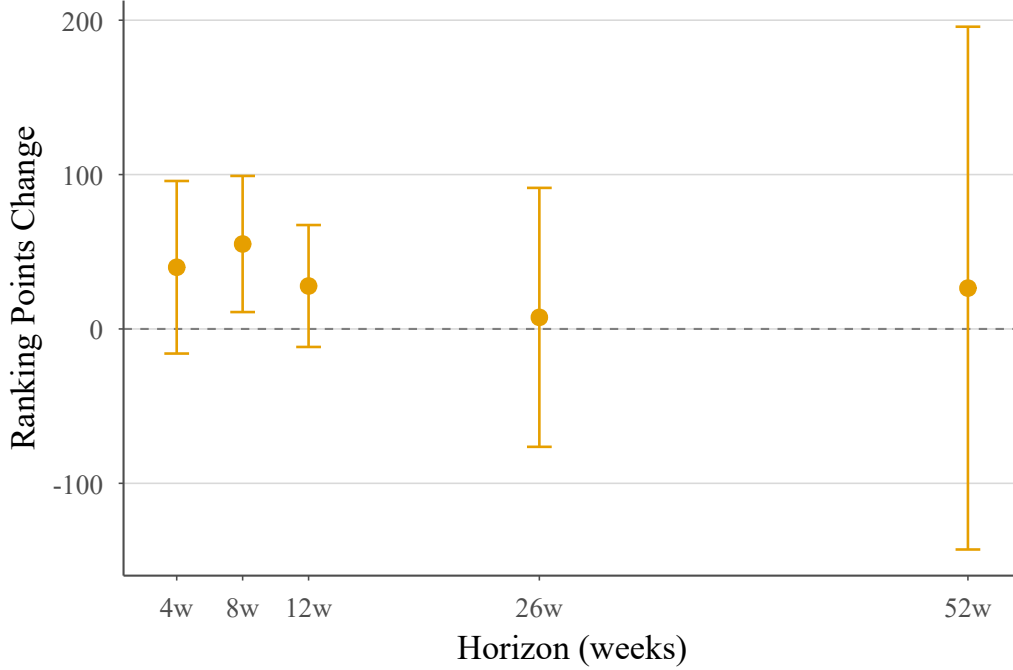


Figure 2: WTA Event Study: Ranking Trajectories for Selected and Unselected Players

Notes. Each point is the estimated treatment effect at the indicated horizon. First-time WTA Grand Slam sample ($N = 82$).

5.3 Heterogeneous Effects

The first-time GS sample is small ($N = 120$ ATP, $N = 82$ WTA), limiting statistical power for subgroup analysis. Table 4 reports heterogeneity for the ATP GS sample. Effects are stronger and more persistent for older players (age ≥ 24): ranking-point gains of 154.9 at 26 weeks ($p < 0.001$) and 186.4 at 52 weeks ($p < 0.01$). Two readings fit this pattern and the data do not separate them. Older players at the qualifying margin have more tournament experience and a denser set of established entry routes, so they may convert a ranking boost into sustained access more efficiently. Alternatively, a player still at the qualifying margin in their late twenties faces a shorter remaining runway and may respond more sharply to an opportunity that is unlikely to recur. The first is a capacity story and the second a motivation story; distinguishing them would require variation in expected career length that this design does not supply. Other subgroup contrasts, by pre-episode ranking and by surface, are imprecisely estimated in the small ATP sample.

The non-GS samples provide substantially more power for heterogeneity analysis ($N = 2,054$ ATP, $N = 1,423$ WTA); results are reported in Appendix Tables 21 and 22.

Table 4: ATP Grand Slam: Heterogeneous Effects

	4w	8w	12w	26w	52w
<i>Rank</i>					
Ranking Pts Δ	71.5** (33.5)	76.5** (30.7)	91.4*** (28.1)	79.6* (44.5)	26.1 (75.4)
$\times$ High	0.6 (42.9)	-10.3 (39.7)	-18.7 (39.8)	-15.6 (47.1)	-6.1 (73.4)
Main Draws	-0.7 (0.6)	-0.4 (0.6)	-0.2 (0.6)	0.8 (0.6)	1.3 (1.7)
$\times$ High	1.2 (0.8)	0.9 (0.8)	0.8 (0.8)	-0.3 (0.9)	-1.0 (2.1)
Elo Δ	-8.8 (9.5)	0.6 (10.5)	-4.3 (10.7)	22.3 (16.5)	-3.7 (25.0)
$\times$ High	8.1 (14.5)	-1.4 (15.2)	-1.5 (16.2)	-34.7* (19.1)	-12.2 (25.9)
<i>Age</i>					
Ranking Pts Δ	76.1** (32.8)	56.6 (34.3)	71.6** (33.8)	6.1 (40.2)	-49.0 (68.3)
$\times$ High	-9.9 (37.3)	35.8 (38.4)	27.8 (39.2)	154.9*** (47.2)	186.4** (74.3)
Main Draws	-0.2 (0.6)	-0.3 (0.6)	-0.3 (0.6)	-0.4 (0.6)	-1.3 (1.7)
$\times$ High	0.1 (0.7)	0.9 (0.7)	1.1* (0.6)	2.5*** (0.8)	5.2** (2.1)
Elo Δ	-9.7 (11.5)	-7.1 (13.0)	-14.7 (12.9)	-13.6 (15.4)	-19.5 (24.1)
$\times$ High	13.9 (15.2)	18.6 (15.7)	24.5 (17.3)	45.7** (19.2)	23.7 (25.5)

Notes. Stacked specification with horizon-specific LL treatment effects estimated by subgroup. First-time ATP Grand Slam sample ($N = 120$), 2006–2024. Low/High rank points split at the tour-specific median. Younger/Older split at age 24. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.4 Non-Grand-Slam Extension

The non-Grand-Slam control-function estimates provide corroborating evidence from substantially larger samples. On the ATP tour ($N = 2,054$), ranking-point gains are 20.6 at 4

weeks ($p = 0.041$) and 44.6 at 26 weeks ($p = 0.002$). Elo effects emerge at medium horizons: +12.8 at 8 weeks ($p = 0.018$) and +30.8 at 26 weeks ($p = 0.016$), a pattern that differs from the GS null and may reflect either the larger sample’s power to detect moderate ability effects or a difference in the non-GS competitive environment.

On the WTA tour ($N = 1,423$), ranking-point gains are 23.2 at 12 weeks ($p = 0.057$), with Elo gains of 10.5 at 12 weeks ($p = 0.011$). The WTA non-GS Elo finding parallels the marginal WTA GS Elo result and points toward genuine, if modest, ability gains, plausibly linked to greater competitive parity on the women’s tour.

The non-GS estimates align with the GS lottery estimates in both direction and magnitude, supporting the validity of the control-function approach. Full non-Grand-Slam dynamic tables are reported in Appendix Tables 19 and 20; heterogeneity results are in Appendix Tables 21 and 22.

The dose-response analysis examines whether treatment effects vary with the number of main-draw matches won at the focal event. The initial LL slot provides a ranking-point benefit that is amplified when the player wins matches in the main draw. Dose-response results are reported in the appendix.

6 Mechanisms

The results establish that lucky loser entry causes gains in ranking points and tournament access. This section decomposes the effect into two channels and examines match-level performance conditional on opponent strength. Throughout, ability means measured skill as captured by Elo: a rating that updates after each match according to the outcome and the two players’ prior ratings. Elo is sensitive to the schedule of opponents faced and to the choice of K -factor, so it proxies for playing strength rather than measuring it directly, and Section 6.1 tests the schedule-sensitivity concern rather than assuming it away.

6.1 Access versus Measured Skill

The career effects of LL entry operate primarily through tournament access rather than improved measured skill. On the ATP tour, ranking points rise significantly through 26 weeks while Elo ratings are null at every horizon. Because Elo adjusts for opponent quality, the null Elo result indicates that treated players do not perform better or worse per match once opponent strength is held constant. The ranking-point gains come entirely from additional tournament participation: LL entry provides a main-draw appearance that generates ranking points, which improve the player’s position in the rolling 52-week ranking, which crosses direct-entry thresholds for subsequent lower-tier events. The evidence aligns with this access cascade: treated players enter more main draws and play more matches at 250+ events over 26 weeks, and their quality-adjusted match performance is unchanged.

The WTA tour behaves differently. Elo rises at 8 weeks (+15.0, $p = 0.082$) and 52 weeks (+36.8, $p = 0.055$) in the pooled Grand Slam sample and at 12 weeks in the non-Grand-Slam sample (+10.5, $p = 0.011$). On the verified-lottery subsample the point estimates keep the same sign and comparable magnitude at short horizons (+12.7 at 4 weeks, +22.1 at 8 weeks) but none reaches significance, which is what the minimum detectable effects for a 41-observation subsample imply. The evidence for a WTA ability channel therefore rests on the pooled and non-Grand-Slam samples, and the verified subsample neither strengthens nor undermines it.

Elo can rise for a reason unrelated to ability, because it rewards wins against any opponent: if the ranking gain routes treated players into lower-tier events with weaker fields, they win more often and Elo increases without any change in how well they play. This schedule-composition account has a direct empirical implication, namely that treated players should face measurably weaker opponents after the focal event. Table 5 tests it by computing, for every match a player contests in the 26 weeks following the event, the opponent’s pre-match Elo, and regressing the player’s mean opponent Elo on treatment.

The data provide no support for the account, and the point estimates run against it.

Treated WTA players face opponents who are 22.9 Elo points *stronger* in the pooled sample and 15.9 points stronger on the verified subsample, where the schedule story requires them to be weaker. Win rates are higher rather than lower (+0.037 pooled, +0.037 verified). Neither difference is statistically significant ($p = 0.121$ pooled, $p = 0.527$ verified), so this is a sign check rather than a rejection, and the verified subsample in particular is too small to resolve differences of this size. What can be said is that the most direct available test gives no encouragement to the schedule account, while the match-level model does not independently confirm the ability account either ($\hat{\delta} = +0.146$, $p = 0.29$). The WTA ability channel therefore remains the more natural reading of the Elo evidence without being established by it.

On the ATP tour the same test returns opponent-strength estimates that are indistinguishable from zero (+22.4, $p = 0.161$ pooled; +13.5, $p = 0.508$ verified) alongside win rates that are flat to slightly lower (−0.017 and −0.037). Treated ATP players play a comparable schedule and do not win more of it, which is what a pure access effect implies.

One caveat applies to the test itself. Mean opponent Elo is computed over the matches a player actually contests in the following 26 weeks, and the composition of that match set is itself affected by treatment: access to higher-tier draws mechanically raises the average opponent’s rating. The test therefore cannot fully separate the two accounts, and a positive coefficient is consistent with either improved play or greater access. Reading it as evidence against the schedule account relies on the additional observation that win rates do not fall, which an access-only story would predict once opponents are stronger.

The access cascade is self-limiting. As the 52-week rolling window progresses, the points from the original LL event constitute a shrinking fraction of the player’s total. Unless downstream tournaments generate enough additional points to replace the original ones, the ranking converges back toward the level determined by sustained match performance. The dynamic results align with this: ATP ranking-point gains peak at 12 weeks and fade toward zero at 52 weeks.

Table 5: Post-Event Schedule Composition and Performance

Sample	Tour	Mean opp. Elo	Matches played	Win rate
Pooled	ATP	22.42 (15.86)	-2.09 (2.44)	-0.017 (0.031)
Pooled	WTA	22.93 (14.61)	-3.07 (2.96)	0.037 (0.030)
Verified	ATP	13.50 (20.24)	-3.05 (3.16)	-0.037 (0.034)
Verified	WTA	15.88 (24.85)	-0.31 (2.78)	0.037 (0.053)

Notes. Outcomes are computed over all matches contested in the 26 weeks following the focal event. Mean opponent Elo averages the opponent’s pre-match Elo rating across those matches. Specification includes Grand Slam $\times$ Year fixed effects and pre-treatment controls, with standard errors clustered at the player level. Under a schedule-composition account of the WTA Elo gains, the coefficient on mean opponent Elo should be negative. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.2 Match-Level Performance

To test directly whether LL entry affects competitive performance itself, I estimate a match-level logit of win probability conditional on opponent strength and tournament context (see Appendix A.1). At Grand Slams, the lottery provides exogenous variation; at non-GS events, a control function corrects for endogenous selection.

First-time ATP GS LL recipients have $\hat{\delta} = -0.141$ (SE = 0.103, $p = 0.17$; player-clustered), a point estimate consistent with overseeding when novice entrants face opponents above their ability level, though not significant at conventional levels. WTA GS recipients have $\hat{\delta} = +0.146$ (SE = 0.138, $p = 0.29$), also not statistically distinguishable from zero. The GS point estimates line up with the direction the overseeding channel predicts, negative for ATP and positive for WTA, but the small Grand Slam samples do not deliver enough power for individually significant tests.

The non-Grand-Slam ATP estimate is $\hat{\delta} = +0.609$ ($p < 0.001$), which reverses the Grand Slam sign and deserves comment rather than assimilation. Three differences separate the two settings. Selection is the first: non-Grand-Slam slots go to the highest-ranked qualifying

losers by rule, so treated players there are positively selected on precisely the ability the outcome measures, and the control function corrects for that selection only under the joint-normality assumption that Section 7.8 shows fails on the ATP randomized benchmark. The estimate is accordingly better read as an upper bound than as a comparable treatment effect. The second is the opponent distribution: a non-Grand-Slam main draw at the 250 or 500 level places a qualifying-margin player against opponents far closer to their own level than a Grand Slam draw does, so the over seeding channel that pushes the Grand Slam estimate negative is weak or absent. The third is composition: the non-Grand-Slam sample is seventeen times larger and spans a wider range of tournament tiers. Given the failed validation, I do not treat the positive non-Grand-Slam estimate as evidence against the Grand Slam over seeding pattern.

Full results including dose and horizon interactions are reported in Appendix Table 23.

The event study figures (Figures 1 and 2) provide a visual summary: ranking-point trajectories diverge sharply after the LL event while pre-event trends are parallel, so the treatment effect is concentrated in post-event access gains rather than pre-existing ability differences.

6.3 Economic Magnitude: Prize Money

Ranking points are an imperfect proxy for the economic stakes. To translate the results into dollars I use publicly available tournament prize tables, and I keep two conceptually distinct quantities separate throughout, because they answer different questions and differ by an order of magnitude.

The first quantity is the *immediate prize*, which is the dollar value of the treatment dose itself. A Grand Slam main-draw entrant earns between \$100,000 and \$400,000 depending on the round reached, against a roughly \$40,000 stipend for a final-round qualifying loser. Averaged over the observed distribution of rounds reached, LL entry raises focal-event earnings by \$93,800 on the ATP tour and \$86,700 on the WTA tour. This is mechanical: it is what

the tournament pays for occupying a main-draw slot, conditional on advancing as far as the player did. It is not a causal cascade, and it would accrue to any player placed in that slot.

The second quantity is the *access cascade*, which is the causal object this paper estimates. Treated players enter roughly 0.66 additional main draws over the following 26 weeks; valued at an average forward-looking first-round prize of \$11,000, that is \$7,300 in downstream earnings on both Grand Slam samples. On the non-Grand-Slam samples, where the immediate prize is far smaller, the cascade dominates the total: \$18,800 of the \$27,800 ATP effect and \$10,200 of the \$16,200 WTA effect come from downstream access rather than the focal event. Appendix Table 35 reports the two components and their sum for all four samples.

The comparison is informative in both directions. A Grand Slam LL slot is worth about \$94,000 on the spot, roughly one to two years of a qualifying-level player’s typical \$40,000–\$80,000 annual earnings, which is why these slots matter enormously to the players who receive them. The downstream cascade adds a further \$7,300 over 26 weeks, about 8 percent of the immediate windfall. Under a $\pm 20\%$ band on the per-entry prize assumption (\$8,800–\$13,200 rather than \$11,000) the cascade component ranges from \$5,800 to \$8,700, and from \$15,000 to \$22,600 on the non-GS ATP sample. The ranking-point cascade this paper identifies is therefore economically real but modest next to the direct payment for the slot, and it fades within the year.

6.4 Reconciling with the Existing Literature

Maity et al. (2025) report positive correlations between lucky loser entry and career outcomes. The match-level tournament performance model, which exploits the lottery for identification, does not detect an improvement in match-level win probability on the ATP tour and reports only a marginal positive effect on the WTA tour. The causal estimates here indicate that the correlational finding in Maity et al. (2025) likely reflects the additional tournament participation that LL entry generates rather than an improvement in competitive performance itself.

The initial conditions literature reports persistent effects of early-career shocks (Oyer, 2006; Oreopoulos et al., 2012). The present results align on the access dimension: LL entry causes persistent changes in the number and type of tournaments a player enters; the ability dimension is largely null. Market transparency is the plausible moderator: in professional tennis, where performance is measured match by match and rankings update weekly, ability reasserts itself within a year. The access channel alone accounts for the ranking-point gains and tournament-entry effects documented in Section 5.

7 Robustness

7.1 Marginal Impact by LL Experience

The main analysis restricts to first-time LL candidates. A natural question is whether subsequent LL opportunities have similar effects, or whether the marginal impact diminishes with experience. To test this, I return to the full (unrestricted) estimation sample and stratify by the number of prior LL awards: $n = 0$ (first LL, matching the main results), $n = 1$ (second LL), and $n \geq 2$ (third or more). Within each experience group, I compare treated players ($D_{ie} = 1$) to controls ($D_{ie} = 0$) who have the same number of prior LL awards, using the full pre-treatment control specification.

Tables 6 and 7 report stratum-by-stratum estimates. These are descriptive: each stratum is estimated on its own subsample, the Grand Slam $n = 1$ and $n \geq 2$ cells are small (ATP $N = 48$ and $N = 80$; WTA $N = 31$ and $N = 19$), and the resulting point estimates are unstable, in places exceeding the physically possible treatment dose in absolute value. Section 7.2 estimates the experience gradient on the pooled sample in a single specification, which is the basis for the paper’s claims about how the return varies with n .

Table 6: Marginal Impact by LL Experience: Grand Slam Samples

Tour	Experience	4w	8w	12w	26w	52w
ATP	First LL ($n = 0$)	72.5***	72.1***	82.8***	72.6**	23.5
		(26.7)	(26.6)	(26.1)	(36.2)	(59.1)
		-4.4	0.2	-4.8	4.7	-9.7
		(9.1)	(10.9)	(10.3)	(13.5)	(21.0)
		-0.2	0.1	0.1	0.7	0.8
		(0.5)	(0.5)	(0.5)	(0.5)	(1.5)
ATP	Second LL ($n = 1$)	0.3	0.7	0.9	1.5	1.3
		(1.0)	(1.0)	(0.9)	(1.1)	(2.8)
		-220.9***	-215.3***	-195.6***	-153.7**	-202.1**
		(48.5)	(44.9)	(49.0)	(64.1)	(98.6)
		-97.9***	-89.9***	-96.5***	-83.7***	-98.6***
		(17.6)	(18.3)	(21.3)	(23.0)	(26.7)
ATP	Third+ LL ($n \geq 2$)	-1.6**	-1.7***	-1.2**	0.6	1.3
		(0.6)	(0.5)	(0.5)	(0.6)	(1.9)
		-9.5***	-9.3***	-8.4***	-4.5***	-2.9
		(1.4)	(1.1)	(1.2)	(1.5)	(3.9)
		-30.7	-38.3	-40.4	4.3	-14.8
		(40.6)	(35.8)	(35.7)	(41.6)	(62.6)
ATP	Third+ LL ($n \geq 2$)	-8.3	-15.0	-18.6	-13.4	-26.5
		(15.2)	(13.6)	(14.3)	(20.5)	(20.1)
		0.4	0.7	0.8	0.5	0.7
		(0.8)	(0.7)	(0.7)	(0.8)	(1.7)
		0.9	1.0	0.9	0.7	1.8
		(1.4)	(1.2)	(1.2)	(1.3)	(2.6)
WTA	First LL ($n = 0$)	40.0	55.0**	27.8	7.5	26.5
		(28.5)	(22.5)	(20.1)	(42.8)	(86.4)
		13.3*	15.0*	11.8	21.6*	36.8*
		(7.6)	(8.5)	(9.1)	(12.7)	(18.9)
		-0.1	0.3	0.3	0.7	1.8
		(0.5)	(0.5)	(0.6)	(0.7)	(1.5)
WTA	Second LL ($n = 1$)	-0.5	0.8	0.6	1.9*	3.2
		(1.0)	(0.7)	(0.8)	(1.0)	(2.5)
		54.6	33.0	-82.8**	-154.7***	-181.3*
		(61.9)	(53.2)	(38.9)	(49.4)	(104.2)
		-6.1	-16.8	-17.9	-16.3	-52.4
		(10.0)	(10.0)	(14.8)	(18.8)	(41.5)
WTA	Second LL ($n = 1$)	-0.6	-0.2	-0.5	-0.7	-2.4
		(1.2)	(1.2)	(1.1)	(1.3)	(2.2)
		-1.0	-0.4	-0.9	-1.7	-6.0
		(2.1)	(2.0)	(1.9)	(2.1)	(3.9)

Notes. Full (unrestricted) Grand Slam sample stratified by number of prior LL awards (n). Within each group, treated = received LL, control = eligible but not selected, both with the same number of prior awards. Event FE specification with full pre-treatment controls (including prior-LL count variables). Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Marginal Impact by LL Experience: Non-Grand-Slam Samples

Tour	Experience	4w	8w	12w	26w	52w
ATP	First LL ($n = 0$)	20.1**	19.7*	13.5	44.5***	37.7
		(10.2)	(10.4)	(10.3)	(14.5)	(30.4)
		6.8	11.1*	10.0*	28.0**	30.4
		(6.3)	(5.7)	(6.0)	(12.7)	(20.5)
		-1.4***	-0.9***	-0.3*	1.7***	5.2***
		(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
		-2.6***	-1.7***	-0.7**	3.0***	9.0***
		(0.4)	(0.4)	(0.3)	(0.5)	(1.2)
		6.1	11.4	2.6	19.6	-1.1
		(24.1)	(26.1)	(28.1)	(34.7)	(53.4)
ATP	Second LL ($n = 1$)	-3.5	-1.5	-1.5	-6.2	-10.9
		(7.9)	(8.6)	(9.1)	(10.4)	(14.3)
		-1.3***	-0.8**	-0.2	1.5***	4.2***
		(0.4)	(0.4)	(0.4)	(0.5)	(1.0)
		-2.4***	-1.4*	-0.6	2.7***	7.4***
		(0.8)	(0.8)	(0.8)	(1.0)	(2.0)
		11.6	11.6	15.5	19.1	-25.5
		(22.8)	(22.7)	(22.9)	(25.5)	(43.2)
		4.2	2.8	3.7	-2.0	-2.3
		(7.1)	(7.4)	(7.8)	(9.4)	(13.3)
ATP	Third+ LL ($n \geq 2$)	-0.9**	-0.4	-0.1	1.3***	3.2***
		(0.4)	(0.3)	(0.3)	(0.4)	(1.0)
		-1.4*	-0.6	0.0	2.4***	5.9***
		(0.7)	(0.7)	(0.7)	(0.9)	(1.9)
WTA	First LL ($n = 0$)	17.8	18.8*	23.0*	20.5	59.2
		(11.3)	(11.0)	(12.2)	(17.9)	(39.6)
		6.5*	7.3*	10.8***	6.4	5.7
		(3.6)	(3.8)	(4.1)	(5.8)	(8.1)
		-1.1***	-0.7***	-0.3**	0.9***	2.7***
		(0.2)	(0.1)	(0.1)	(0.2)	(0.5)
		-2.1***	-1.4***	-0.8***	1.6***	5.6***
		(0.3)	(0.3)	(0.3)	(0.4)	(1.1)
		36.6	29.6	34.4	20.9	-18.0
		(27.9)	(29.3)	(29.3)	(36.4)	(65.0)
WTA	Second LL ($n = 1$)	8.0	5.2	6.2	4.9	9.4
		(6.5)	(6.9)	(7.4)	(9.5)	(11.8)
		-1.0***	-0.6	-0.1	1.4***	3.4***
		(0.4)	(0.4)	(0.4)	(0.5)	(0.9)
		-1.6**	-1.0	-0.0	2.6***	6.2***
		(0.7)	(0.7)	(0.7)	(0.9)	(1.8)
	Third+ LL ($n \geq 2$)	17.2	-1.5	-17.1	-81.4**	-159.6**
		(26.3)	(29.0)	(29.8)	(38.9)	(72.1)
		3.3	1.4	3.6	-1.4	-24.9*
		(6.4)	(7.1)	(6.9)	(10.0)	(15.0)
		-0.4	-0.0	0.4	1.1***	2.4**
		(0.3)	(0.3)	(0.3)	(0.4)	(1.1)
		-1.0	-0.6	-0.4	0.5	2.7
		(0.7)	(0.7)	(0.7)	(1.0)	(2.1)

Notes. Full (unrestricted) non-Grand-Slam sample stratified by number of prior LL awards (n). Control function specification with $\hat{v}_{ie} \times$ horizon terms and full pre-treatment controls. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.2 The Experience Gradient

Comparing coefficients estimated on separate stratum subsamples does not identify how the return varies with experience, because each subsample carries its own covariate distribution and its own residual variance. A pooled regression coefficient is a variance-weighted average of stratum effects whose implicit weights depend on the within-stratum variance of treatment conditional on covariates, and those weights need not match the population share of each stratum (Słoczyński, 2022). Differences between a pooled coefficient and a share-weighted average of subsample coefficients are therefore uninformative about heterogeneity.

I instead estimate the gradient in a single saturated specification on the pooled sample:

$$\begin{aligned}
Y_{ie,h} = & \alpha_e + \alpha_h + \sum_h \beta_h D_{ie} \mathbf{1}[h] \\
& + \sum_h \gamma_{1h} D_{ie} \mathbf{1}[h] \mathbf{1}[n_{ie} = 1] + \sum_h \gamma_{2h} D_{ie} \mathbf{1}[h] \mathbf{1}[n_{ie} \geq 2] \\
& + \sum_h \lambda_{1h} \mathbf{1}[h] \mathbf{1}[n_{ie} = 1] + \sum_h \lambda_{2h} \mathbf{1}[h] \mathbf{1}[n_{ie} \geq 2] \\
& + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h},
\end{aligned} \tag{6}$$

where n_{ie} counts prior LL awards. Here β_h is the effect for first-time recipients, $\beta_h + \gamma_{1h}$ the effect for second-time recipients, and $\beta_h + \gamma_{2h}$ the effect for third-plus recipients. The λ_{jh} terms let each experience stratum carry its own untreated outcome path across horizons; without them the specification would force LL-naïve and repeat candidates to share a common untreated trajectory, and any difference would be absorbed into $\hat{\gamma}_{jh}$ and misread as treatment-effect heterogeneity. All coefficients come from one regression on one sample, so they aggregate to the pooled estimand by construction and the question becomes a hypothesis test rather than an accounting identity.

The specification must be saturated on the control side as well as the treatment side. Without stratum-specific horizon effects, the model forces the untreated outcome path to share a common horizon profile across experience strata, and since players who reach $n \geq 2$

have by construction spent years losing in final qualifying, their untreated ranking trajectory differs from a first-timer's. Any such difference would be absorbed into $\hat{\gamma}_{2h}$ and misread as a treatment shifter. Equation (6) therefore includes $\mathbf{1}[n = 1]$ and $\mathbf{1}[n \geq 2]$ interacted with the horizon indicators, so each stratum carries its own untreated path and $\hat{\gamma}_{jh}$ is identified from the treated-versus-control contrast within stratum.

Table 8 reports the results, and they support a weaker claim than the paper's original framing supposed. On the ATP Grand Slam sample the first-time effect is 55.2 ranking points at 4 weeks, 58.4 at 8 weeks, and 69.1 at 12 weeks, each significant at the 1% level. The second-LL shifter is small and statistically indistinguishable from zero at every horizon (-6.7 , 3.7 , and 55.4 at 4, 12, and 26 weeks, none approaching significance), so a player receiving a second lucky loser slot gains about as much as a player receiving a first. The third-plus shifter is negative and significant at the three short horizons (-79.7 , $p = 0.014$; -87.4 , $p = 0.011$; -99.3 , $p = 0.007$), enough to offset the base effect, but it is indistinguishable from zero at 26 and 52 weeks.

Crucially, the joint Wald test does *not* reject a flat experience profile for ranking points on any of the four samples (ATP Grand Slam $F = 1.20$, $p = 0.285$; non-Grand-Slam ATP $F = 0.54$, $p = 0.861$; WTA Grand Slam $F = 1.21$, $p = 0.281$; non-Grand-Slam WTA $F = 1.08$, $p = 0.374$). Only main draws entered on the ATP Grand Slam sample rejects ($F = 2.15$, $p = 0.019$). The honest summary is therefore that the return to an opportunity shows no statistically established gradient in the number already received, with the qualified exception of short-horizon ranking points and main-draw access on the ATP Grand Slam sample, where third-plus recipients do worse.

What the evidence does rule out is the specific claim that the *first* opportunity is distinctive. The second-LL shifter is small and insignificant on every sample and at every horizon, and the implied second-LL effects on the ATP Grand Slam sample (48.5 , 52.8 , 72.8 at 4, 8, and 12 weeks) track the first-LL effects closely. Whatever gradient exists does not begin between the first opportunity and the second. The restriction to first-time recipients in the

main analysis remains the right sample choice for estimating the effect of an access shock cleanly, but it should not be read as isolating a uniquely pivotal event.

Two caveats bound this evidence. First, n_{ie} is itself an outcome of prior treatment: within a stratum the lottery randomizes D , so β_h and $\beta_h + \gamma_{jh}$ are each valid effects for their own population, but the comparison across strata is between endogenously constituted groups, and the joint test is a test of equality across treatment-history-defined populations rather than of a causal dose-response. Second, an earlier version of this specification omitted the stratum-by-horizon terms and produced a $\hat{\gamma}_{2h}$ that grew monotonically with the horizon, reaching -182.3 at 52 weeks and implying third-plus effects larger in absolute value than the physically feasible treatment dose. That pattern was the signature of a differential untreated path rather than a treatment effect, and it disappears once the control side is saturated.

7.3 Verified Lottery Subsample and Contamination

The Grand Slam lottery mechanism operates only when withdrawals occur before qualifying completes; post-qualifying withdrawals follow the ranking rule (the highest-ranked qualifying loser receives the slot). Withdrawal timing is not recorded in the data, so the main sample may pool lottery-assigned and ranking-assigned observations. This contamination has a signable direction: under ranking-based assignment, treated players are systematically higher-ranked than controls, biasing ranking-point effect estimates upward rather than toward attenuation. Two features of the analysis partially offset this bias. First, the pre-treatment control vector Z_{ie}^{pre} includes ranking points (linear and quadratic), Elo, and surface Elo, absorbing the observable component of selection. Second, within-event fixed effects restrict comparisons to the top-four pool at a given tournament, where the ranking gap between treated and untreated candidates is mechanically small (at most three ranks).

To assess the residual contamination concern, I construct a *verified lottery* subsample using two criteria, neither of which requires external information. Under the ranking rule an event that fills k slots awards them to exactly the top k ranked qualifying losers, in rank

Table 8: Experience Gradient: Saturated Interaction Model (Ranking Points)

	4w	8w	12w	26w	52w
<i>Panel: GS-ATP</i>					
$\hat{\beta}_h$ (first LL, $n = 0$)	55.2*** (18.5)	58.4*** (18.7)	69.1*** (20.1)	61.4 (41.1)	4.7 (64.4)
$\hat{\gamma}_{1h}$ (shift, $n = 1$)	-6.7 (38.2)	-5.6 (34.9)	3.7 (37.4)	55.4 (60.6)	70.4 (108.5)
$\hat{\gamma}_{2h}$ (shift, $n \geq 2$)	-79.7** (32.3)	-87.4** (34.2)	-99.3*** (36.9)	-58.4 (58.4)	-21.4 (90.7)
Implied ATT, $n = 1$	48.5	52.8	72.8	116.8	75.2
Implied ATT, $n \geq 2$	-24.5	-29.0	-30.2	2.9	-16.7
<i>Panel: GS-WTA</i>					
$\hat{\beta}_h$ (first LL, $n = 0$)	38.8 (27.6)	54.8** (25.5)	27.9 (24.7)	3.6 (43.7)	18.3 (88.5)
$\hat{\gamma}_{1h}$ (shift, $n = 1$)	111.1* (63.1)	59.3 (65.7)	-29.6 (65.9)	-71.7 (85.5)	-108.3 (138.7)
$\hat{\gamma}_{2h}$ (shift, $n \geq 2$)	-61.4 (96.9)	6.0 (93.5)	20.0 (102.3)	128.7 (114.9)	82.7 (186.5)
Implied ATT, $n = 1$	149.9	114.1	-1.7	-68.0	-90.0
Implied ATT, $n \geq 2$	-22.6	60.7	47.9	132.3	101.0
<i>Panel: NonGS-ATP</i>					
$\hat{\beta}_h$ (first LL, $n = 0$)	4.2 (8.4)	6.0 (9.6)	1.0 (10.1)	34.7** (14.3)	28.9 (28.6)
$\hat{\gamma}_{1h}$ (shift, $n = 1$)	13.3 (11.0)	14.4 (13.7)	18.6 (15.2)	5.4 (23.8)	-14.1 (46.9)
$\hat{\gamma}_{2h}$ (shift, $n \geq 2$)	6.1 (10.5)	0.4 (11.7)	0.5 (12.5)	-27.6 (20.3)	-69.1* (41.0)
Implied ATT, $n = 1$	17.5	20.4	19.5	40.1	14.8
Implied ATT, $n \geq 2$	10.3	6.4	1.4	7.1	-40.3
<i>Panel: NonGS-WTA</i>					
$\hat{\beta}_h$ (first LL, $n = 0$)	23.0** (10.4)	21.2** (10.8)	30.7** (12.6)	22.9 (18.4)	47.5 (39.2)
$\hat{\gamma}_{1h}$ (shift, $n = 1$)	-4.1 (13.5)	-3.6 (15.6)	-18.1 (19.0)	-17.3 (33.7)	-51.2 (65.4)
$\hat{\gamma}_{2h}$ (shift, $n \geq 2$)	17.4 (13.7)	4.8 (16.8)	-13.0 (19.0)	-67.8** (30.6)	-152.6** (68.2)
Implied ATT, $n = 1$	19.0	17.5	12.6	5.6	-3.8
Implied ATT, $n \geq 2$	40.4	26.0	17.7	-44.9	-105.2

Notes. Estimates of equation (6) on the full unrestricted sample. $\hat{\beta}_h$ is the treatment effect for first-time recipients ($n = 0$); $\hat{\gamma}_{1h}$ and $\hat{\gamma}_{2h}$ shift that effect for second-time ($n = 1$) and third-plus ($n \geq 2$) recipients. Implied ATTs add the shifter to the base effect; their standard errors use the delta method and account for the covariance between $\hat{\beta}_h$ and $\hat{\gamma}_{jh}$. Grand Slam panels use Grand Slam $\times$ Year and horizon fixed effects; non-Grand-Slam panels add the control-function residual interacted with horizon. Pre-treatment controls include prior-LL counts, which vary within the pooled sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

order, and each criterion identifies a distinct observable violation of that pattern.

The first is individual. A treated player whose rank among final-round qualifying losers exceeds the event’s slot count cannot have been placed by the ranking rule, so their entry is provably a lottery draw. The second is at the event level. If k slots were filled by rank order, the set of recipient ranks is exactly $\{1, 2, \dots, k\}$; a gap in that set cannot arise under ranking and therefore proves that a lottery ran at that event. The second criterion identifies events the first misses: at a four-slot event whose recipients are ranked first and third, neither rank exceeds four, but the ranking rule cannot skip rank two. The gap test must be computed on the unrestricted sample rather than the first-time-restricted one, because repeat recipients excluded by the sample restriction would leave holes in the rank set that mimic genuine gaps. Both criteria share the assumption that no higher-ranked loser declined a slot, since a decline would also produce a gap.

Applying both criteria, 41 of the 64 Grand Slam events with first-time candidates are certified as lotteries, yielding 26 verified ATP events with $N = 64$ observations (35 treated, 29 control) and 15 verified WTA events with $N = 41$ observations (23 treated, 18 control).

Table 9 reports stacked dynamic estimates on this subsample. On the ATP tour the ranking-point coefficients keep the sign and the rising shape of the pooled estimates, running from 26.5 at 4 weeks to 87.5 at 26 weeks, but the standard errors are half again as large as in the pooled sample and only the 26-week estimate approaches significance ($p = 0.055$). On the WTA tour the 8-week gain survives at +70.7 ($p = 0.044$), close to the pooled estimate of 55.0. Elo is null at every horizon on both tours except a marginally negative ATP estimate at 52 weeks (-51.3 , $p = 0.074$).

These results neither confirm nor refute the headline estimates, and the reason is power rather than contradiction. Table 15 reports minimum detectable effects: at 80% power the verified ATP subsample detects roughly 90 to 125 ranking points over the 4-to-26-week range, against pooled point estimates of 72 to 83. The subsample is not capable of resolving an effect of the size the pooled sample identifies, so its nulls carry little information and

elevating it to primary would substitute an underpowered null for a powered estimate. Nor can it corroborate them. The verified ATP point estimates at short horizons (26.5, 37.4, 41.6) run 50 to 64 percent below the pooled values, which is the direction upward contamination would produce; the differences are inside one verified standard error, so nothing is established either way, but the pattern should not be described as reassuring. A sample that cannot distinguish 72 from zero equally cannot distinguish 72 from 26.

An earlier and narrower version of this subsample, built on the individual rank criterion alone, produced a smaller sample ($N = 49$ ATP) on which Elo appeared to decline sharply at 12 and 52 weeks and WTA Elo appeared to rise at four of five horizons. Adding the event-level gap criterion enlarges the verified sample by roughly a third and those apparent effects attenuate to insignificance. The larger sample is the better-identified one, since the gap criterion certifies genuine lotteries that the narrower rule missed, so the attenuated estimates supersede the earlier ones. The episode illustrates how fragile inference is at these sample sizes and is the reason the paper rests its contamination defence on the placebo tests of Section 7.5 rather than on this subsample.

The verified subsample is therefore uninformative in both directions, and the defence of the pooled estimates against contamination rests on the placebo tests of Section 7.5 rather than on this subsample. Those tests target the contamination mechanism directly, since ranking-rule assignment selects on pre-treatment rank, and they do so at full sample size.

A partial news audit supports the classification rule, and its coverage should be stated precisely. Of the 65 first-time Grand Slam entries whose rank falls within the slot count, I searched tennis-media coverage and tournament records for 23: those ranked exactly first among qualifying losers, which is the subset where a ranking-rule assignment is most likely a priori, since that rule always fills the first vacancy with the top-ranked loser. Ten of the 23 were classified as ranking-rule assignments, seven of them with a specific withdrawal date and three on the basis of reported late-injury withdrawals without an exact date. Every one was a post-qualifying withdrawal, which triggers the ranking rule: Kyrgios withdrew from

Wimbledon on the Sunday night before the 2023 main draw, Alexandrova from Wimbledon hours before her first-round match in 2024, Davidovich Fokina from Wimbledon the same year, Raonic from Roland Garros in 2021 and the US Open in 2019, Andreescu and Badosa from the US Open in 2023, and Tomljanovic from the Australian Open in 2023. None was a lottery draw. Thirteen were searched without yielding a datable source and remain unresolved; these are predominantly pre-2015 events with thin contemporaneous coverage. Appendix Table 33 lists every searched entry with the withdrawn player, the date where one could be established, and the classification reached.

The remaining 42 ambiguous entries, ranked second through fourth at events offering two to seven slots, have not been audited. They are classified as ambiguous on the rank criterion alone and are excluded from the verified subsample on that basis. This is the conservative treatment, because an unaudited entry is never counted as a verified lottery: auditing them can only move entries into the verified subsample or confirm their exclusion. The verified-lottery sample sizes reported above are therefore a lower bound on what complete verification would yield, and the corresponding estimates are correspondingly less precise than they would be under a full audit. Extending the audit to these 42 entries is the most direct available route to a larger verified subsample.

7.4 Pre-Treatment Balance

Under genuine within-pool randomization, balance holds in expectation. Finite-sample imbalance is inevitable: with only four candidates per pool and a binary treatment, any single draw produces deviations that are properly interpreted as sampling variability rather than as evidence against randomization. Controls included in the outcome specification improve precision by reducing residual variance; they do not correct for a violation of random assignment, because under the lottery no such violation is present in expectation.

Table 10 reports formal balance tests on pre-treatment covariates, and the two tours look different enough that they warrant separate discussion.

Table 9: First-LL Verified Lottery Subsample: Dynamic Effects

	4w	8w	12w	26w	52w
<i>ATP: N = 64 (35 treated, 29 control), 26 events</i>					
Ranking Pts Δ	26.5 (34.5)	37.4 (34.2)	41.6 (32.9)	87.5* (44.6)	29.1 (69.0)
Elo Δ	-11.5 (11.6)	-11.7 (14.8)	-16.7 (13.8)	8.1 (18.1)	-51.3* (28.2)
Main Draws	-0.25 (0.67)	-0.35 (0.62)	-0.08 (0.63)	0.73 (0.72)	1.47 (1.79)
Matches 250+	-0.70 (1.24)	-0.70 (1.16)	-0.09 (1.16)	1.68 (1.32)	2.17 (3.40)
<i>WTA: N = 41 (23 treated, 18 control), 15 events</i>					
Ranking Pts Δ	67.2 (45.9)	70.7** (33.9)	24.1 (35.2)	-19.5 (66.8)	-24.2 (127.0)
Elo Δ	12.7 (11.3)	22.1 (13.2)	17.8 (14.7)	5.2 (18.5)	26.9 (30.6)
Main Draws	0.50 (0.97)	0.80 (0.98)	1.02 (1.06)	0.89 (1.17)	1.02 (2.25)
Matches 250+	0.11 (1.84)	1.29 (1.72)	1.32 (1.78)	1.93 (2.05)	1.34 (4.07)

Notes. Restricted to events with at least one LL entry whose rank among final-round qualifying losers exceeds the number of available LL slots at that event, which the ranking rule cannot produce. First-time LL candidates only. Stacked dynamic specification with Grand Slam $\times$ Year fixed effects, horizon fixed effects, and pre-treatment controls. Standard errors clustered at the player level. Minimum detectable effects for this subsample are reported in Table 15. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

On the ATP tour the picture is close to what randomization implies. The joint F -test yields $p = 0.106$, ranking points and Elo are balanced to three decimal places on the Elo measures, and the one covariate rejecting at the 5% level is age ($p = 0.034$), where treated players average 24.9 years against 26.2 for controls, a gap of 1.3 years within a pool spanning 18 to 35. Age cannot plausibly be correlated with a lottery draw and enters Z_{ie}^{pre} in any case. The within-event test yields $p = 0.046$, which with seven covariates across two tours is within the range chance produces.

The WTA tour is a different matter and the paper should not gloss it. Four of seven covariates reject at the 5% level, and all four run in the same direction: treated players carry more ranking points (0.53 against 0.42, $p = 0.007$; the squared term $p = 0.006$) and a higher Elo (18.54 against 17.42, $p = 0.022$; squared term $p = 0.022$). The two pairs are collinear by construction, so this is roughly two independent dimensions rather than four, but both are precisely the dimensions on which ranking-rule assignment would select. The joint F -test does not reject at conventional levels ($p = 0.063$), and the within-event test is far from rejecting ($p = 0.516$), which suggests the imbalance operates between events rather than within the pools where the lottery runs. That is the reading most consistent with the design, since event fixed effects absorb between-pool variation, and it is supported by the pre-treatment placebo tests of Section 7.5, which are null on the WTA tour at all three horizons. It is not, however, a reading the balance table alone establishes. The honest position is that the WTA lottery sample shows level imbalance on exactly the covariates contamination would move, that the within-event and placebo evidence argues against this reflecting non-random assignment within pools, and that the WTA estimates should be read with more caution than the ATP estimates on that account. Section 5.2 already treats the WTA ranking-point result as suggestive rather than established on independent grounds, and this balance evidence reinforces that reading.

Event study plots (Figures 1 and 2) show parallel trajectories in the weeks preceding each Grand Slam. Non-GS balance tests are reported in the appendix; that sample is not

randomized, so imbalance there is expected and motivates the control function.

Table 10: Pre-Treatment Balance: Grand Slam Samples

Variable	ATP			WTA		
	LL	Control	p	LL	Control	p
Ranking points / 1000	0.48	0.45	0.088	0.53	0.42	0.007
(Ranking points / 1000) ²	0.25	0.21	0.083	0.31	0.20	0.006
Elo / 100	17.30	17.30	0.977	18.54	17.42	0.022
(Elo / 100) ²	299.27	299.28	0.977	343.91	303.37	0.022
Surface Elo / 100	16.29	16.38	0.686	17.62	17.28	0.407
(Surface Elo / 100) ²	266.91	270.07	0.676	314.28	301.26	0.366
Age	24.93	26.20	0.034	23.15	23.83	0.391
Joint F -test p -value	0.106			0.063		
N	120			82		

Notes. Columns compare means for treated (LL selected) and control (unselected) players at the player-episode level. Covariates include ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and age. Prior LL variables excluded (zero by construction for first-time candidates). Joint F -statistic tests the null that all covariates are jointly balanced. First-time ATP: $N = 120$; first-time WTA: $N = 82$. Grand Slam sample, 2006–2024.

7.5 Pre-Treatment Placebo Tests

Balance on levels and visually parallel event-study trajectories are suggestive, but the direct test of random assignment is whether eventual treatment status predicts outcomes measured *before* treatment. I construct ranking-point changes over the twelve, eight, and four weeks preceding each focal event, using the same range-join to the weekly ranking files that produces the post-treatment outcomes, and regress each on eventual LL status with the headline specification.

Table 11 reports the results. None of the twelve tests, spanning three pre-treatment horizons, two tours, and both the pooled and verified-lottery samples, rejects at the 5% level; the largest absolute t -statistic in the set is 1.47. Point estimates are small relative to the post-treatment effects: on the pooled ATP sample they run from -4.3 to $+5.9$ ranking points, against post-treatment effects of 72 to 83, and they alternate in sign rather than trending. This is the sharpest available evidence against the residual contamination concern. Ranking-

rule assignment selects the highest-ranked qualifying loser, so contaminated observations would carry pre-treatment ranking trajectories that differ systematically between treated and control players. No such difference appears.

Table 11: Pre-Treatment Placebo Tests					
Sample	Tour		-12w	-8w	-4w
Pooled	ATP	$\hat{\beta}$	-3.0 (20.7)	5.9 (15.5)	-4.3 (10.3)
Pooled	WTA	$\hat{\beta}$	40.0 (27.1)	22.1 (20.8)	-15.0 (11.8)
Verified	ATP	$\hat{\beta}$	10.1 (23.8)	-2.2 (18.5)	4.5 (16.6)
Verified	WTA	$\hat{\beta}$	-14.6 (51.9)	-2.3 (41.7)	-22.1 (25.2)

Notes. Outcome is the change in ranking points from the indicated pre-treatment week to the focal event date, so all variation predates assignment. Specification matches the headline dynamic model with Grand Slam $\times$ Year fixed effects and pre-treatment controls. Standard errors clustered at the player level. Under random assignment these coefficients should be indistinguishable from zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.6 Attrition and Lee Bounds

Players leave the rankings, and outcomes are observed only for those who remain. If treatment itself keeps players ranked, long-horizon outcomes condition on a treatment-affected selection event. Table 12 reports observation rates. Through 26 weeks the rates are close: 80.3% of treated and 78.0% of control ATP players are observed at 26 weeks, a gap of 2.4 percentage points that a two-proportion test cannot distinguish from zero ($p = 0.93$). At 52 weeks the gap widens to 12.7 percentage points on the ATP tour (75.4% versus 62.7%), and reverses to -10.0 points on the WTA tour; neither is individually significant at conventional levels given the sample sizes ($p = 0.19$ and $p = 0.39$), but both are large enough to matter for interpretation.

Table 13 reports Lee (2009) bounds, which trim the over-observed group from each tail of its outcome distribution to equalize observation rates and so bound the effect for the subpopulation observed under either assignment. Two features of the table govern how much it can be asked to carry. The quantity bounded is the raw difference in mean ranking-point change between treated and control players, not the regression-adjusted coefficients reported in Tables 2 and 3; the two differ because the headline specification absorbs Grand Slam $\times$ Year effects and pre-treatment covariates that the raw difference does not. And where observation rates are already equal, no trimming occurs and the bounds collapse to a point, which is what produces the degenerate WTA entries at 12 and 26 weeks.

On the ATP tour the bounds are tight and positive through 26 weeks: the 12-week raw difference of 44.4 ranking points is bounded within $[38.3, 50.1]$, and the 26-week difference of 37.2 within $[23.4, 46.8]$. Differential survival cannot generate the short- and medium-horizon ATP result. The WTA tour supports no such statement. Its 4- and 8-week bounds are positive ($[23.8, 23.8]$ and $[32.4, 44.1]$), but the 12- and 26-week entries are degenerate points of 10.8 and -11.5 , the latter negative, so the WTA evidence at medium horizons is that there is nothing to bound rather than that a positive effect survives bounding. At 52 weeks the bounds widen sharply and span zero on both tours ($[-143.9, 35.3]$ ATP; $[-18.2, 90.7]$ WTA), because the trimming fraction rises to 17% and 12% respectively. The paper’s fade-out claim at 52 weeks is consequently the one result that differential attrition can also explain, and Section 8 reports it with that caveat rather than as established.

7.7 Multiple Testing

The paper reports four outcomes at five horizons on four samples, and the resulting 80 tests invite a correction. Which correction is appropriate depends on what is being controlled, and the three standards reported here give materially different answers, so I state all of them rather than selecting the most favourable.

Within the ranking-point outcome, treating the five horizons as one family of 20 tests,

Table 12: Attrition: Outcome Observation Rates by Treatment Status

Tour		4w	8w	12w	26w	52w
ATP	Treated (%)	100.0	91.8	86.9	80.3	75.4
	Control (%)	100.0	94.9	89.8	78.0	62.7
	Difference	0.0	-3.1	-2.9	2.4	12.7
WTA	Treated (%)	100.0	97.0	97.0	93.9	75.8
	Control (%)	100.0	100.0	98.0	91.8	85.7
	Difference	0.0	-3.0	-1.0	2.1	-10.0

Notes. Share of players with a non-missing ranking-point outcome at each horizon. Difference row reports treated minus control in percentage points, with significance from a two-proportion test. First-time Grand Slam samples. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 13: Lee (2009) Bounds on the Ranking-Point Effect

Tour		4w	8w	12w	26w	52w
ATP	Unadjusted	22.4	27.7	44.4	37.2	-42.0
	Lee lower	22.4	23.4	38.3	23.4	-143.9
	Lee upper	22.4	32.3	50.1	46.8	35.3
WTA	Unadjusted	23.8	35.6	10.8	-11.5	19.2
	Lee lower	23.8	32.4	10.8	-11.5	-18.2
	Lee upper	23.8	44.1	10.8	-11.5	90.7

Notes. Unadjusted row reports the raw difference in mean ranking-point change between treated and control players among those observed. Lee bounds trim the over-observed group from each tail of its outcome distribution until observation rates are equal, bounding the effect for the always-observed subpopulation under monotone selection. Bounds coincide with the unadjusted difference where observation rates are already equal. First-time Grand Slam samples.

the ATP Grand Slam effects at 4, 8, and 12 weeks survive Benjamini-Hochberg correction ($q = 0.039, 0.039, 0.020$); the 26-week effect does not ($q = 0.119$), and the WTA 8-week effect is marginal ($q = 0.067$). Applying Benjamini-Hochberg jointly across all 80 tests gives a similar picture: 23 tests survive at $q < 0.05$, including the three ATP Grand Slam horizons ($q = 0.028, 0.028, 0.009$).

Romano-Wolf step-down inference is stricter and gives a different answer. Where Benjamini-Hochberg bounds the expected share of rejections that are false, Romano-Wolf bounds the probability of making any false rejection at all, and it does so using a null-imposed wild cluster bootstrap that preserves the dependence among the test statistics (Romano and Wolf, 2005; Clarke et al., 2020). Of the 80 tests, 16 survive at the 5% level, and all 16 are non-Grand-Slam access outcomes: main draws entered and matches at 250-level events, at 4, 8, 26 and 52 weeks on both tours, where the raw p -values fall below 0.001. No Grand Slam ranking-point effect survives. On the ATP tour the step-down p -values are 0.348 at 4 weeks, 0.348 at 8 weeks, and 0.130 at 12 weeks; the WTA 8-week effect returns 0.558.

The right way to read this is not that the Grand Slam results are spurious, but that family-wise error control across 80 tests is a demanding standard that a single design with 120 observations was never going to meet. The dynamic path across horizons is one economic claim rather than five independent ones, and the appropriate correction for a pre-specified primary outcome is the within-outcome family, which the Grand Slam effects clear. What the Romano-Wolf results do establish is a ranking of evidential strength. The non-Grand-Slam access effects are robust to the strictest available correction and are the most secure findings in the paper. The Grand Slam ranking-point effects are supported by randomization inference, by placebo tests, and by attrition bounds, and they survive false-discovery-rate control, but they are not individually strong enough to clear family-wise control over the full outcome set. Readers who require the latter standard should treat the Grand Slam estimates as suggestive and the non-Grand-Slam access results as established.

7.8 Control Function Specification Check

The CF specification check (Appendix Table 27) applies the control function to the randomized GS sample using a uniform selection probability $P_i^{LL} = c_e/N_e$. On the WTA tour, the specification passes: $\hat{\rho}_h$ is insignificant at all horizons for ranking points (e.g., $\hat{\rho}_4 = 17.5$, SE 48.5) and treatment effect estimates change modestly. On the ATP tour, the specification fails: $\hat{\rho}_h$ is large and highly significant (e.g., $\hat{\rho}_4 = 90.6$, $p < 0.01$), indicating that the uniform probability is a poor approximation for the ATP GS assignment process. This likely reflects that the ATP GS pool size varies substantially across events (from 4 to 10+ candidates depending on the qualifying draw size and withdrawal timing), so the uniform-within-top-4 assumption mischaracterizes the selection. The ATP GS failure does not compromise the lottery-based estimates, which rely only on random assignment within the top-four pool rather than on a functional form assumption. It does indicate that the non-GS CF estimates should be read as corroboration rather than as a free-standing identification strategy, and that the lottery design is the primary basis for causal inference.

7.9 Wildcards

Wildcard entries provide a parallel mechanism for main-draw access. If control players differentially receive wildcards to subsequent events, the counterfactual is contaminated. In the first-LL sample, wildcards are rare among qualifying-level players (ranked 100th–300th), who are generally below the threshold for wildcard consideration at tour-level events. In the 26-week window following the focal event, the average number of wildcard main-draw entries is 0.08 per treated player and 0.09 per control player on the ATP tour ($p = 0.71$), and 0.12 versus 0.11 on the WTA tour ($p = 0.83$). Differential wildcard receipt is therefore too small to confound the LL treatment effect.

7.10 Small-Sample Inference

The Grand Slam samples are small, so asymptotic cluster-robust inference warrants scrutiny. Table 14 reports observation counts, treated and control counts, player clusters, and events for every sample. Because players rarely appear in the pool more than once under the first-time restriction, cluster counts track sample sizes closely: 109 players in the pooled ATP sample, 77 in the pooled WTA sample, and 62 and 40 in the verified-lottery ATP and WTA subsamples. The verified WTA subsample sits at the boundary of the range where asymptotic clustered inference is normally considered reliable, so results on both verified subsamples are reported alongside bootstrap p -values below.

Table 15 reports minimum detectable effects at 80% power and a 5% two-sided test, computed as $(z_{0.975} + z_{0.80})$ times the realised clustered standard error. These convert null results into statements about what the data can and cannot rule out, and the comparison across samples is stark. On the pooled ATP sample the minimum detectable ranking-point effect is 74.8 at 4 weeks, 74.4 at 8 weeks, and 73.1 at 12 weeks, bracketing the estimated effects of 72.5, 72.1, and 82.8. At 4 and 8 weeks the minimum detectable effect slightly exceeds the estimate; only at 12 weeks does it fall below. The design has approximately the power required to detect effects of the size it reports, and no margin beyond that. On the verified-lottery ATP subsample the minimum detectable effect rises to 92 to 125 over the same range while the point estimates fall, so a null there is close to uninformative. The pattern is sharper still on the WTA side, where the verified subsample requires 95 to 129 points at short horizons.

Two implications follow. First, the paper is operating at the edge of its power throughout, and results should be read with the corresponding caution: this is a design with 120 and 82 observations, not one that can resolve modest effects. Second, and for the same reason, the null results on the verified subsample carry little evidential weight against the pooled estimates, since that subsample could not have detected the pooled effect had it been present.

Table 16 reports wild cluster bootstrap p -values following [Cameron et al. \(2008\)](#), using

the restricted bootstrap with the null imposed and Webb six-point weights, which outperform Rademacher weights when the cluster count is small. The bootstrap re-estimates the full specification 9,999 times per coefficient, and is computed with the `fwildclusterboot` implementation of [Roodman et al. \(2019\)](#).

The bootstrap corroborates the asymptotic inference rather than overturning it, and it tracks the asymptotic p -values closely at every horizon on every sample. On the pooled ATP sample the bootstrap values at 4, 8, and 12 weeks are 0.006, 0.009, and 0.002 against asymptotic values of 0.008, 0.008, and 0.002. On the pooled WTA sample the 8-week bootstrap value is 0.010 against 0.017, and on the verified WTA subsample it is 0.040 against 0.044. One coefficient crosses the conventional threshold: the 26-week ATP effect moves from $p = 0.047$ to $p_{\text{boot}} = 0.052$. Everywhere else the two procedures agree on every rejection and every failure to reject. Cluster-robust standard errors are therefore not materially over-rejecting here, which is consistent with the cluster counts in [Table 14](#) sitting above the range where the small-cluster problem is acute.

The 26-week ATP effect is accordingly the weakest of the estimates the paper calls significant, and the three inference procedures line up against it two to one. It clears the 5% threshold under clustered standard errors ($p = 0.047$) and fails under both the wild bootstrap ($p_{\text{boot}} = 0.052$) and Fisher randomization inference ($p_{\text{Fisher}} = 0.103$). The gap between the bootstrap and the randomization test is itself informative about what each procedure assumes. Both bootstrap and asymptotic inference treat the regression as correctly specified and resample from it, whereas the Fisher test permutes the treatment indicator within the blocks in which it was actually assigned and so tests the sharp null using the design itself. For a lottery, randomization inference is the more appropriate standard, and the 26-week result fails it by the widest margin of the three procedures. The paper reports the estimate but does not rest any claim on it.

Table 14: Sample and Cluster Counts

Sample	N	Treated	Control	Players	Events
GS-ATP (pooled first-LL)	120	61	59	109	55
GS-WTA (pooled first-LL)	82	33	49	77	32
GS-ATP (verified lottery)	64	35	29	62	26
GS-WTA (verified lottery)	41	23	18	40	15
NonGS-ATP (first-LL)	2054	523	1531	830	513
NonGS-WTA (first-LL)	1423	378	1045	736	422

Notes. Player clusters are the unit at which standard errors are clustered throughout. Events count distinct tour $\times$ tournament $\times$ year combinations contributing identifying variation.

Table 15: Minimum Detectable Effects (Ranking Points, 80% Power)

Sample		4w	8w	12w	26w	52w
Pooled-ATP	$\hat{\beta}_h$	72.5	72.1	82.8	72.6	23.5
	MDE (80%)	74.8	74.4	73.1	101.4	165.5
Verified-ATP	$\hat{\beta}_h$	26.5	37.4	41.6	87.5	29.1
	MDE (80%)	96.6	95.8	92.2	125.0	193.2
Pooled-WTA	$\hat{\beta}_h$	40.0	55.0	27.8	7.5	26.5
	MDE (80%)	79.9	63.0	56.4	119.9	242.0
Verified-WTA	$\hat{\beta}_h$	67.2	70.7	24.1	-19.5	-24.2
	MDE (80%)	128.7	95.0	98.6	187.0	355.9

Notes. Minimum detectable effect at 80% power for a 5% two-sided test, computed as $2.80 \times \text{SE}$ using the realised player-clustered standard error from the stacked dynamic specification. A null result is informative only where the minimum detectable effect lies below the effect size of interest.

Table 16: Wild Cluster Bootstrap Inference (Ranking Points)

Sample		4w	8w	12w	26w	52w
Verified-ATP	$\hat{\beta}_h$	26.5	37.4	41.6	87.5	29.1
	p (clustered)	0.444	0.279	0.211	0.055	0.675
	p (wild boot.)	0.453	0.319	0.263	0.076	0.671
Verified-WTA	$\hat{\beta}_h$	67.2	70.7	24.1	-19.5	-24.2
	p (clustered)	0.151	0.044	0.498	0.771	0.850
	p (wild boot.)	0.161	0.040	0.482	0.817	0.856
Pooled-ATP	$\hat{\beta}_h$	72.5	72.1	82.8	72.6	23.5
	p (clustered)	0.008	0.008	0.002	0.047	0.692
	p (wild boot.)	0.006	0.009	0.002	0.052	0.700
Pooled-WTA	$\hat{\beta}_h$	40.0	55.0	27.8	7.5	26.5
	p (clustered)	0.165	0.017	0.171	0.861	0.760
	p (wild boot.)	0.145	0.010	0.156	0.882	0.776

Notes. Restricted wild cluster bootstrap with the null imposed, Webb six-point weights, and 9,999 replications, clustered at the player level (Cameron et al., 2008), computed with the `fwildclusterboot` implementation of Roodman et al. (2019). Clustered p -values are the asymptotic values reported elsewhere in the paper.

8 Conclusion

A randomly assigned tournament entry raises a marginal player’s ranking for several months, and the gain operates through access to further tournaments rather than through improved play. In the first-time ATP Grand Slam sample ($N = 120$), lottery selection raises ranking points by 72.5 at 4 weeks, 72.1 at 8 weeks, and 82.8 at 12 weeks, each surviving Fisher randomization inference at 5,000 permutations ($p_{\text{Fisher}} = 0.006, 0.022, 0.025$). The 26-week estimate of 72.6 clears the threshold under clustered inference ($p = 0.047$) but under neither the wild cluster bootstrap ($p_{\text{boot}} = 0.052$) nor Fisher ($p_{\text{Fisher}} = 0.103$), and the effect is gone by 52 weeks. On the WTA tour ($N = 82$) the 8-week gain of 55.0 clears clustered but not randomization inference ($p = 0.017$ against $p_{\text{Fisher}} = 0.077$), so the WTA access result is the weaker of the two. Non-Grand-Slam control-function estimates on larger samples ($N = 2,054$ ATP; $N = 1,423$ WTA) reproduce the ATP access pattern, and these non-Grand-Slam access effects are the only results in the paper that survive family-wise error control across the full 80-test outcome set.

Two features of the design carry most of the weight in supporting a causal reading. Pre-treatment placebo regressions at -12 , -8 , and -4 weeks return coefficients indistinguishable from zero in all twelve tests across both tours and both the pooled and verified-lottery samples, which is what random assignment implies and what ranking-rule assignment would not produce. Lee (2009) bounds that correct for differential attrition are tight and stay positive through 26 weeks on the ATP tour, so the short- and medium-horizon ATP results are not an artifact of treated players remaining observable at higher rates. The bounds are computed on the raw treated-control difference rather than the regression-adjusted coefficient, and on the WTA tour they support no analogous claim beyond 8 weeks.

The mechanism is access accumulation. The ranking system does not adjust for opponent quality: a first-round Grand Slam win earns the same points regardless of who was beaten. Points earned at the focal event improve the player’s position in the rolling ranking, which grants entry to subsequent tournaments and further chances to accumulate points. Elo, which does adjust for opponent quality, is flat on the ATP tour at every horizon, so the ranking gains reflect more draws entered and more matches played rather than better play. The evidence on the ATP tour also points weakly toward over seeding, in the sense that a player whose ranking runs ahead of their ability draws stronger opponents: the match-level estimate is negative but imprecise ($\hat{\delta} = -0.141$, $p = 0.17$), while the verified-lottery subsample shows an Elo decline only at 52 weeks. The WTA evidence differs, with Elo rising at several horizons in a pattern that a schedule-composition account can also generate.

The return to an opportunity does decline with the number already received, but not between the first and the second. In a saturated specification estimated on the pooled sample, the second-LL shifter is statistically indistinguishable from zero at every horizon, while the third-plus shifter is large, negative, and significant throughout, and a joint test rejects a flat profile on three of four samples. Returns are stable across the first two opportunities and collapse at the third, which is what a threshold account predicts when the direct-entry cutoff sits above what a single main-draw appearance delivers.

The self-limiting character of the cascade is the finding that travels furthest. Where [Oyer \(2006\)](#) and [Oreopoulos et al. \(2012\)](#) document effects of initial conditions that run for a decade, the access gain here is gone inside a year. The difference is not the size of the shock but the memory of the market. Tennis rankings are a rolling 52-week window: the points that generated the advantage are mechanically retired on a known schedule, and sustained performance is the only thing that replaces them. That suggests a way to classify tournament labor markets by how long an access shock should survive in each. Markets that continuously measure performance and periodically reset the record, such as sales tournaments with quarterly quotas or ranked freelance platforms with recency-weighted ratings, should show decay on the timescale of the reset. Markets that measure performance coarsely and record it permanently, such as academic tenure clocks, partnership tracks, and other up-or-out promotion ladders, have no comparable retirement mechanism, and an early access shock there should persist in the way the initial-conditions literature finds. [Pallais \(2014\)](#) sits at the persistent end: an oDesk rating, once earned, does not expire. The prediction is testable wherever a market’s record-keeping horizon can be measured, and it implies that the durability of an opportunity shock is a design parameter of the market rather than a fact about workers.

The economic magnitudes separate into two quantities that should not be added casually. A Grand Slam main-draw slot pays about \$94,000 more than a qualifying exit, which is one to two years of a qualifying-level player’s typical earnings and explains why these slots matter so much to the players who receive them. That figure is mechanical: it is what the tournament pays for the slot. The causal cascade this paper identifies, the additional main draws entered over the following 26 weeks, is worth roughly \$7,300, about 8 percent of the immediate windfall. On non-Grand-Slam events the ordering reverses and the cascade dominates.

Three limitations bound the results. The Grand Slam lottery samples are small, and the verified-lottery subsample that restricts to provably randomized events is smaller still

($N = 64$ ATP, $N = 41$ WTA), so nulls on that subsample cannot separate a zero effect from an effect the sample lacks the power to detect. Lottery contamination remains partially unresolved: withdrawal timing is not in the public record, two data-only criteria certify 44 of 64 events as lotteries, and a news audit of a subset of the remainder dated ten entries, all to post-qualifying withdrawals that trigger the ranking rule. Finally, the 52-week fade-out is the one result that differential attrition can also explain, since treated players remain in the rankings at higher rates at that horizon and the Lee bounds widen to include zero. Longer-run outcomes, and settings where the market's record-keeping horizon differs, are the natural next tests.

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A Econometric Details

A.1 Match-Level Model

The match-level probability is specified as:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta D_{ie} + \rho v_{ie}) \quad (7)$$

where X_{ijm} contains pairwise covariates (ranking points difference / 1000 linear and quadratic, Elo difference / 100 linear and quadratic, surface Elo difference / 100 linear and quadratic, Laplace-smoothed H2H win proportion, H2H encounter count, age difference, and surface indicators) and Z_{ie}^{pre} contains pre-treatment focal player characteristics (ranking points / 1000 linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, and player age). The trajectory likelihood factorizes into match-level Bernoulli contributions, so estimation reduces to a standard logit on stacked match-level data.

A.2 Control Function

The generalized residual correcting for endogenous LL assignment at non-Grand-Slam events is:

$$\hat{v}_{ie} = D_{ie} \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{P_i^{LL}} - (1 - D_{ie}) \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{1 - P_i^{LL}} \quad (8)$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal PDF and CDF. Including $\hat{v}_{ie}$ in the outcome equation corrects for selection on unobservables.

A.3 Selection Probability Construction

The selection probability P_i^{LL} is computed from the full combinatorial structure of qualifying match pairings. A weighted logit of match outcomes is estimated on qualifying round matches

at non-Grand-Slam LL-granting events, separately for each tour. Given predicted match-winning probabilities for the $N_e - 1$ other final-round qualifying matches, P_i^{LL} is computed by exact enumeration over all 2^{N_e-1} outcome combinations. When $N_e - 1 > 20$, Monte Carlo simulation with 10,000 draws approximates the probability. For the first-LL-restricted sample, the win model is re-estimated on the restricted event set (ATP: $N = 9,425$ qualifying matches; WTA: $N = 10,009$), and the selection probability is recomputed for the LL-naive pool at each event. The old-vs-new P_i^{LL} correlation is 0.979, confirming stability.

A.4 Inference for the Match-Level Model

Standard errors for the match-level model are cluster-robust at the player level, allowing arbitrary within-player dependence across matches and events. Fisher randomization inference for the ranking-point outcomes (Section 5) is reported alongside the clustered SEs and is preferred for the small Grand Slam samples ($N = 120$ ATP; $N = 82$ WTA).

B Additional Tables and Figures

Table 17: Summary Statistics: ATP Grand Slam (First-LL)

Variable	LL Selected			Not Selected		
	Mean	SD	<i>N</i>	Mean	SD	<i>N</i>
<i>Panel A: Pre-Treatment</i>						
Age	24.93	3.41	61	26.20	3.08	59
Ranking points	482.69	121.82	61	447.24	103.05	59
Elo rating	17.35	0.67	61	17.35	0.67	59
Surface Elo	1629.07	124.59	61	1638.44	128.59	59
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	16.93	62.61	61	-5.44	54.73	59
Ranking points Δ (26w)	83.86	228.02	49	46.67	281.37	46
Ranking points Δ (52w)	45.46	316.89	46	87.46	361.27	37
Elo Δ (4w)	-0.24	25.19	60	0.17	29.63	57
Elo Δ (26w)	22.73	62.54	41	10.37	64.76	43
Elo Δ (52w)	14.66	76.14	40	20.66	87.12	32
Main draws entered (26w)	4.69	3.42	52	4.00	2.71	48
Main draws entered (52w)	10.18	8.07	50	9.54	7.28	41

Notes. First-time LL candidates at ATP Grand Slams, 2006–2024. *N* = 120 (61 treated, 59 control). All players have zero prior LL awards.

Table 18: Summary Statistics: WTA Grand Slam (First-LL)

Variable	LL Selected			Not Selected		
	Mean	SD	<i>N</i>	Mean	SD	<i>N</i>
<i>Panel A: Pre-Treatment</i>						
Age	23.15	3.75	33	23.83	3.16	49
Ranking points	525.70	179.38	33	420.84	147.57	49
Elo rating	19.16	1.13	33	18.59	0.97	49
Surface Elo	1762.44	194.16	33	1728.13	163.39	49
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	45.67	68.34	33	21.88	39.93	49
Ranking points Δ (26w)	114.65	222.06	31	126.13	235.35	45
Ranking points Δ (52w)	168.92	432.09	25	149.71	297.64	42
Elo Δ (4w)	7.04	25.40	32	3.61	23.57	48
Elo Δ (26w)	16.51	54.66	24	5.94	54.32	38
Elo Δ (52w)	34.80	82.30	23	4.60	67.33	42
Main draws entered (26w)	4.84	3.03	31	3.89	2.84	47
Main draws entered (52w)	11.90	7.50	29	9.78	6.61	45

Notes. First-time LL candidates at WTA Grand Slams, 2006–2024. $N = 82$ (33 treated, 49 control).

Table 19: Non-Grand-Slam Dynamic Effects: ATP (First-LL, Control Function)

	4w	8w	12w	26w	52w
Ranking Pts Δ	20.6** (10.1)	20.5** (10.3)	14.1 (10.1)	44.6*** (14.2)	41.1 (29.9)
Elo Δ	8.4 (6.2)	12.8** (5.4)	11.4** (5.6)	30.8** (12.8)	33.6* (20.3)
Main Draws	-1.4*** (0.2)	-0.9*** (0.2)	-0.3* (0.2)	1.7*** (0.3)	5.2*** (0.6)
Matches 250+	-2.6*** (0.4)	-1.7*** (0.3)	-0.6* (0.3)	3.0*** (0.5)	8.9*** (1.1)
Win %	-0.04 (0.03)	-0.01 (0.02)	-0.00 (0.03)	0.04 (0.03)	0.03 (0.03)

Notes. Control function estimates. First-time ATP non-GS sample ($N = 2,054$, 523 treated), 2007–2024. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 20: Non-Grand-Slam Dynamic Effects: WTA (First-LL, Control Function)

	4w	8w	12w	26w	52w
Ranking Pts Δ	18.1 (11.4)	19.2* (11.1)	23.2* (12.2)	19.2 (17.8)	58.5 (40.1)
Elo Δ	6.2* (3.6)	7.1* (3.8)	10.5** (4.1)	6.2 (5.9)	5.3 (8.2)
Main Draws	-1.1*** (0.2)	-0.7*** (0.1)	-0.4*** (0.1)	0.9*** (0.2)	2.7*** (0.5)
Matches 250+	-2.1*** (0.3)	-1.4*** (0.3)	-0.8*** (0.3)	1.7*** (0.4)	5.6*** (1.1)
Win %	0.01 (0.02)	0.02 (0.02)	0.00 (0.02)	-0.01 (0.02)	-0.01 (0.02)

Notes. Control function estimates. First-time WTA non-GS sample ($N = 1,423$, 378 treated), 2009–2024. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 21: Non-Grand-Slam Heterogeneity: ATP (First-LL)

	4w	8w	12w	26w	52w
<i>Rank</i>					
Ranking Pts Δ	19.0*	19.6*	13.4	46.4***	48.5
	(11.0)	(11.3)	(11.3)	(16.2)	(32.1)
$\times$ High	11.9	7.2	5.0	-8.3	-40.4
	(10.9)	(11.2)	(11.7)	(18.7)	(27.3)
Main Draws	-1.7***	-1.1***	-0.4**	2.0***	6.2***
	(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
$\times$ High	2.1***	1.5***	0.9***	-1.4***	-5.7***
	(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
Elo Δ	8.6	12.9**	11.9**	28.8**	33.2
	(6.2)	(5.6)	(6.0)	(13.0)	(20.3)
$\times$ High	-1.7	-1.5	-4.2	12.0	1.7
	(5.9)	(6.3)	(7.2)	(9.8)	(14.1)
<i>Age</i>					
Ranking Pts Δ	26.9**	28.6**	21.3*	51.6***	46.4
	(11.4)	(12.0)	(11.8)	(16.8)	(31.9)
$\times$ High	-18.2	-23.8*	-21.3	-20.7	-16.7
	(11.2)	(13.0)	(15.1)	(23.2)	(39.6)
Main Draws	-1.4***	-0.9***	-0.3*	1.7***	5.1***
	(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
$\times$ High	-0.0	-0.0	0.1	0.1	0.3
	(0.3)	(0.3)	(0.2)	(0.4)	(1.0)
Elo Δ	13.7**	17.0***	16.5***	30.1**	28.5
	(6.6)	(5.9)	(6.1)	(13.2)	(20.3)
$\times$ High	-16.0***	-12.4**	-15.3**	3.0	18.8
	(5.1)	(5.8)	(6.6)	(8.8)	(12.8)

Notes. Interaction specifications by rank and age. First-time ATP non-GS sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 22: Non-Grand-Slam Heterogeneity: WTA (First-LL)

	4w	8w	12w	26w	52w
<i>Rank</i>					
Ranking Pts Δ	16.5 (14.3)	15.1 (14.3)	22.5 (15.7)	13.0 (22.8)	74.5 (51.4)
$\times$ High	5.8 (14.0)	14.0 (14.3)	2.7 (16.0)	20.4 (23.2)	-57.9 (51.3)
Main Draws	-1.7*** (0.2)	-1.0*** (0.2)	-0.6*** (0.2)	1.5*** (0.3)	4.8*** (0.6)
$\times$ High	2.1*** (0.2)	1.3*** (0.2)	0.7*** (0.2)	-1.7*** (0.3)	-6.4*** (0.6)
Elo Δ	4.2 (3.9)	3.5 (4.2)	6.2 (4.5)	1.3 (6.3)	2.2 (9.3)
$\times$ High	4.9 (4.9)	10.7* (5.4)	12.9** (5.8)	14.1* (8.1)	8.9 (12.4)
<i>Age</i>					
Ranking Pts Δ	14.9 (14.0)	9.1 (14.6)	6.1 (17.0)	-29.1 (22.7)	-34.0 (39.6)
$\times$ High	6.2 (14.2)	18.6 (15.9)	30.6* (18.2)	82.7*** (31.0)	144.4** (58.4)
Main Draws	-1.0*** (0.2)	-0.5*** (0.2)	-0.3* (0.2)	0.9*** (0.3)	1.9*** (0.6)
$\times$ High	-0.2 (0.2)	-0.3 (0.2)	-0.1 (0.2)	0.1 (0.3)	1.3* (0.8)
Elo Δ	7.8* (4.3)	10.5** (4.6)	13.0** (5.0)	7.1 (7.5)	1.1 (11.5)
$\times$ High	-3.0 (4.5)	-6.2 (4.9)	-4.6 (5.4)	-1.9 (7.7)	6.4 (12.7)

Notes. Interaction specifications by rank and age. First-time WTA non-GS sample. Standard errors clustered at the player level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 23: Match-Level Tournament Performance: All Four Samples (First-LL)

	GS-ATP	GS-WTA	NonGS-ATP	NonGS-WTA
$\hat{\delta}$ (LL entry)	-0.141 (0.103)	0.146 (0.138)	0.609*** (0.026)	-0.033 (0.029)
Matches	4,445	3,715	407,959	47,376

Notes. Logit estimates of LL entry on match win probability. Pooled specification with pairwise covariates and focal player controls. First-LL restricted samples. Standard errors clustered at the player level. The non-GS specification includes the control function residual. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 24: Pre-Treatment Balance: Non-Grand-Slam (First-LL)

Variable	ATP			WTA		
	LL	Control	p	LL	Control	p
Ranking points / 1000	0.38	0.25	0.000	0.56	0.39	0.000
(Ranking points / 1000) ²	0.19	0.09	0.000	0.41	0.22	0.000
Elo / 100	17.30	16.97	0.000	18.23	17.58	0.000
(Elo / 100) ²	300.03	289.95	0.000	334.01	311.54	0.000
Surface Elo / 100	16.43	16.33	0.084	18.23	17.87	0.000
(Surface Elo / 100) ²	271.54	267.70	0.053	334.01	321.26	0.000
Age	26.15	25.18	0.000	24.02	23.63	0.108
Joint F -test p -value	0.000			0.000		
N	2054			1423		

Notes. Non-GS sample is not randomized; imbalance is expected and motivates the control function. First-time ATP: $N = 2,054$; First-time WTA: $N = 1,423$.

Table 25: Non-Grand-Slam Immediate Effects (First-LL, Control Function)

Outcome	t -test	CF + Tourney FE	
	LL Mean	$\hat{\beta}$	(SE)
<i>ATP</i>			
Main-draw match win	0.26***	0.39***	(0.02)
Main-draw matches played	1.38***	1.58***	(0.03)
Ranking points at event	0.00	NA	(NA)
N	523 / 1531	2054	
<i>WTA</i>			
Main-draw match win	0.37***	0.38***	(0.02)
Main-draw matches played	1.50***	1.50***	(0.03)
Ranking points at event	0.00	NA	(NA)
N	378 / 1045	1423	
<i>Pooled</i>			
Main-draw match win	0.31***	0.38***	(0.01)
Main-draw matches played	1.43***	1.54***	(0.02)
Ranking points at event	0.00	NA	(NA)
N	901 / 2576	3477	

Notes. Control function estimates at the focal event. First-time LL candidates only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 26: Fisher Randomization Inference (First-LL)

	4w	8w	12w	26w	52w
<i>Panel A: ATP</i>					
Ranking pts Δ	72.532*** (26.699)	72.081** (26.564)	82.806** (26.106)	72.630 (36.211)	23.475 (59.087)
Fisher p	0.006	0.022	0.025	0.103	0.774
Elo Δ	-4.435 (9.070)	0.192 (10.877)	-4.826 (10.271)	4.653 (13.479)	-9.734 (21.014)
Fisher p	0.662	0.986	0.691	0.786	0.681
Main draws entered	-0.160 (0.520)	0.063 (0.484)	0.132 (0.465)	0.663 (0.546)	0.836 (1.457)
Fisher p	0.728	0.896	0.790	0.325	0.507
<i>Panel B: WTA</i>					
Ranking pts Δ	39.955 (28.517)	55.029* (22.498)	27.822 (20.148)	7.508 (42.791)	26.456 (86.391)
Fisher p	0.236	0.077	0.343	0.881	0.803
Elo Δ	13.254 (7.560)	14.966 (8.499)	11.843 (9.057)	21.591 (12.693)	36.840 (18.941)
Fisher p	0.105	0.152	0.298	0.158	0.101
Main draws entered	-0.129 (0.544)	0.274 (0.546)	0.264 (0.581)	0.657 (0.672)	1.751 (1.500)
Fisher p	0.832	0.626	0.689	0.441	0.231
Permutations: 5,000; treatment permuted within slam $\times$ year					
N (ATP)	103 $\times$ 5 horizons				
N (WTA)	77 $\times$ 5 horizons				

Notes. Fisher p -values from 5,000 permutations of the treatment indicator within Grand Slam $\times$ Year blocks. The specification matches the headline dynamic model, with horizon-invariant coefficients on the pre-treatment controls. Permutation retains only blocks containing at least one treated and one control observation, since singleton blocks contribute no permutation variation; this reduces the effective sample from $N = 120$ to $N = 103$ on the ATP tour and from $N = 82$ to $N = 77$ on the WTA tour.

Table 27: Control Function Validation on GS Lottery Sample (First-LL)

	4w	8w	12w	26w	52w
<i>ATP</i>					
Ranking Pts Δ	72.5*** (26.7)	72.1*** (26.6)	82.8*** (26.1)	72.6** (36.2)	23.5 (59.1)
+ CF	-106.2* (57.2)	-103.3* (60.4)	-93.0 (60.0)	-52.3 (61.6)	-128.4 (85.6)
$\hat{\rho}_h$					
Elo Δ	-4.4 (9.1)	0.2 (10.9)	-4.8 (10.3)	4.7 (13.5)	-9.7 (21.0)
+ CF	-89.3*** (21.7)	-82.5*** (23.4)	-87.1*** (23.1)	-80.2*** (26.1)	-124.5*** (34.5)
$\hat{\rho}_h$					
Main Draws	-0.2 (0.5)	0.1 (0.5)	0.1 (0.5)	0.7 (0.5)	0.8 (1.5)
+ CF	-2.2* (1.3)	-1.9 (1.3)	-1.8 (1.3)	-0.9 (1.4)	-0.1 (2.2)
$\hat{\rho}_h$					
<i>WTA</i>					
Ranking Pts Δ	40.0 (28.5)	55.0** (22.5)	27.8 (20.1)	7.5 (42.8)	26.5 (86.4)
+ CF	9.9 (79.9)	18.6 (72.8)	26.5 (73.4)	-6.6 (94.2)	-5.4 (133.5)
$\hat{\rho}_h$					
Elo Δ	13.3* (7.6)	15.0* (8.5)	11.8 (9.1)	21.6* (12.7)	36.8* (18.9)
+ CF	24.1 (21.5)	24.8 (23.2)	21.8 (22.8)	26.9 (25.0)	56.7* (32.5)
$\hat{\rho}_h$					
Main Draws	-0.1 (0.5)	0.3 (0.5)	0.3 (0.6)	0.7 (0.7)	1.8 (1.5)
+ CF	1.6 (1.2)	2.0* (1.2)	2.0 (1.2)	2.4* (1.3)	3.0 (2.4)
$\hat{\rho}_h$					

Notes. CF applied to randomized GS sample. Under correct specification, treatment effects are unchanged and $\hat{\rho}_h \approx 0$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 28: Minimum Detectable Effects at 80% Power (First-LL)

	GS-ATP	GS-WTA	NonGS-ATP	NonGS-WTA
<i>Panel A: Match-level performance (δ model)</i>				
$\hat{\delta}$ (log-odds)	—	—	—	—
SE	—	—	—	—
MDE (log-odds, 80% power)	—	—	—	—
MDE (pp win prob)	—	—	—	—
N matches	—	—	—	—
<i>Panel B: Elo change (stacked dynamic model, 26w horizon)</i>				
$\hat{\beta}_{26w}$ (Elo pts)	4.7	21.6	30.7	6.2
SE at 26w	13.5	12.7	12.8	5.9
MDE at 26w (Elo pts)	37.7	35.5	35.8	16.4
<i>Panel C: MDE by horizon (Elo points)</i>				
4w	25.4	21.2	17.4	10.1
8w	30.5	23.8	15.1	10.7
12w	28.8	25.4	15.7	11.5
26w	37.7	35.5	35.8	16.4
52w	58.8	53.0	56.9	22.9

Notes. MDE = $(z_{0.025} + z_{0.80}) \times \sigma / \sqrt{N \times p \times (1 - p)}$, where p is the proportion treated. First-LL GS samples.

Table 29: Economic Magnitudes (First-LL)

	Control Mean	LL Effect	Effect/Mean	Effect/SD
<i>Panel A: ATP Grand Slam (First-LL)</i>				
Ranking pts (baseline)	447	—	—	—
Elo (baseline)	17	—	—	—
Ranking Pts Δ (4w)	-5.4	72.5***	1333.1%	1.325
Ranking Pts Δ (26w)	46.7	72.6**	155.6%	0.258
Ranking Pts Δ (52w)	87.5	23.5	26.8%	0.065
Elo Δ (4w)	0.2	-4.4	-2570.4%	-0.150
Elo Δ (26w)	10.4	4.7	44.9%	0.072
Elo Δ (52w)	20.7	-9.7	-47.1%	-0.112
Main Draws (4w)	0.9	-0.2	-18.5%	-0.195
Main Draws (26w)	4.0	0.7	16.6%	0.245
Main Draws (52w)	9.5	0.8	8.8%	0.115
<i>Panel B: WTA Grand Slam (First-LL)</i>				
Ranking pts (baseline)	421	—	—	—
Elo (baseline)	19	—	—	—
Ranking Pts Δ (4w)	21.9	40.0	182.6%	1.001
Ranking Pts Δ (26w)	126.1	7.5	6.0%	0.032
Ranking Pts Δ (52w)	149.7	26.5	17.7%	0.089
Elo Δ (4w)	3.6	13.3*	367.1%	0.562
Elo Δ (26w)	5.9	21.6*	363.2%	0.397
Elo Δ (52w)	4.6	36.8*	800.9%	0.547
Main Draws (4w)	0.6	-0.1	-22.6%	-0.169
Main Draws (26w)	3.9	0.7	16.9%	0.232
Main Draws (52w)	9.8	1.8	17.9%	0.265
<i>Panel C: ATP Non-Grand Slam (First-LL)</i>				
Ranking pts (baseline)	251	—	—	—
Elo (baseline)	1626	—	—	—
Ranking Pts Δ (4w)	2.8	20.5**	719.4%	0.618
Ranking Pts Δ (26w)	15.0	44.5***	296.5%	0.366
Ranking Pts Δ (52w)	10.0	41.0	410.4%	0.207
Elo Δ (4w)	5.4	8.3	153.8%	0.327
Elo Δ (26w)	11.1	30.7**	276.8%	0.509
Elo Δ (52w)	9.1	33.6*	369.0%	0.454
Main Draws (4w)	0.3	-1.4***	-555.8%	-2.440
Main Draws (26w)	1.6	1.7***	105.3%	0.569
Main Draws (52w)	2.8	5.2***	185.4%	0.926
<i>Panel D: WTA Non-Grand Slam (First-LL)</i>				
Ranking pts (baseline)	390	—	—	—
Elo (baseline)	1842	—	—	—
Ranking Pts Δ (4w)	10.0	18.1	181.2%	0.313
Ranking Pts Δ (26w)	32.1	19.2	59.8%	0.108
Ranking Pts Δ (52w)	32.8	58.5	178.2%	0.199
Elo Δ (4w)	2.0	6.2*	302.3%	0.236
Elo Δ (26w)	12.7	6.2	48.5%	0.101
Elo Δ (52w)	17.6	5.3	30.1%	0.066
Main Draws (4w)	0.5	-1.1***	-203.7%	-1.330
Main Draws (26w)	2.6	0.9***	35.8%	0.271
Main Draws (52w)	4.5	2.7***	59.6%	0.426

Notes. Control group means, treatment effects, and effect-to-mean ratios for key outcomes. First-LL samples.

Table 30: Dose-Response: Grand Slam (First-LL)

	4w	8w	12w	26w	52w
<i>GS-ATP</i>					
Ranking Pts Δ	81.8*** (30.9)	81.7*** (29.7)	91.6*** (28.6)	69.6* (40.5)	5.0 (61.0)
$\times$ Dose	-23.3 (27.3)	-24.2 (27.5)	-20.2 (27.3)	17.6 (28.8)	72.6 (60.3)
Main Draws	0.3 (0.6)	0.5 (0.6)	0.4 (0.6)	0.7 (0.7)	0.3 (1.6)
$\times$ Dose	-1.0** (0.4)	-0.8** (0.4)	-0.5 (0.5)	0.1 (0.6)	2.3 (1.6)
Elo Δ	-1.9 (11.6)	4.3 (12.6)	-2.9 (12.5)	2.3 (15.2)	-10.0 (22.1)
$\times$ Dose	-4.8 (9.5)	-9.2 (10.3)	-3.1 (11.3)	16.5 (20.1)	4.0 (20.5)
<i>GS-WTA</i>					
Ranking Pts Δ	38.6 (33.3)	37.1 (29.2)	10.9 (26.7)	-36.8 (45.9)	-51.0 (84.8)
$\times$ Dose	-38.5 (56.2)	23.4 (44.7)	20.0 (43.0)	126.5*** (33.4)	209.4 (155.7)
Main Draws	0.3 (0.7)	0.6 (0.7)	0.7 (0.7)	0.6 (0.8)	1.2 (1.7)
$\times$ Dose	-1.2 (0.9)	-1.2 (0.9)	-1.4 (1.0)	0.4 (1.3)	2.1 (2.0)
Elo Δ	4.7 (11.2)	4.4 (11.5)	-1.5 (11.3)	9.3 (15.6)	32.6 (23.0)
$\times$ Dose	15.4 (14.9)	23.8* (13.5)	39.1** (16.1)	28.4* (16.5)	3.4 (27.7)

Notes. Treatment interacted with matches won at the LL event. First-LL GS samples. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 31: Within-Event Balance Tests: Grand Slam (First-LL)

Variable	ATP		WTA	
	Coef	p	Coef	p
Ranking points / 1000	0.009	0.599	0.022	0.188
(Ranking points / 1000) ²	0.015	0.371	0.023	0.205
Elo / 100	-0.211	0.144	0.108	0.524
(Elo / 100) ²	-7.192	0.151	3.998	0.531
Surface Elo / 100	-0.154	0.497	-0.146	0.574
(Surface Elo / 100) ²	-5.581	0.448	-4.737	0.607
Age	-1.612	0.040	-0.990	0.285

Notes. Each covariate regressed on treatment indicator with event (Grand Slam $\times$ Year) fixed effects. Joint F -test: ATP $p = 0.046$; WTA $p = 0.516$. First-LL sample.

Table 32: Benjamini-Hochberg q -Values Across Primary Outcomes

Sample		4w	8w	12w	26w	52w
<i>Panel: Ranking points</i>						
GS-ATP	$\hat{\beta}_h$	72.5***	72.1***	82.8***	72.6**	23.5
	p	0.008	0.008	0.002	0.047	0.692
	q (BH)	0.028	0.028	0.009	0.126	0.780
GS-WTA	$\hat{\beta}_h$	40.0	55.0**	27.8	7.5	26.5
	p	0.165	0.017	0.171	0.861	0.760
	q (BH)	0.280	0.054	0.280	0.883	0.822
NonGS-ATP	$\hat{\beta}_h$	20.6**	20.5**	14.1	44.6***	41.1
	p	0.041	0.047	0.166	0.002	0.170
	q (BH)	0.119	0.126	0.280	0.008	0.280
NonGS-WTA	$\hat{\beta}_h$	18.1	19.2*	23.2*	19.2	58.5
	p	0.112	0.085	0.057	0.281	0.145
	q (BH)	0.209	0.174	0.135	0.409	0.264
<i>Panel: Elo</i>						
GS-ATP	$\hat{\beta}_h$	-4.4	0.2	-4.8	4.7	-9.7
	p	0.626	0.986	0.639	0.731	0.644
	q (BH)	0.745	0.986	0.745	0.812	0.745
GS-WTA	$\hat{\beta}_h$	13.3*	15.0*	11.8	21.6*	36.8*
	p	0.084	0.082	0.195	0.093	0.055
	q (BH)	0.174	0.174	0.306	0.182	0.134
NonGS-ATP	$\hat{\beta}_h$	8.4	12.8**	11.4**	30.8**	33.6*
	p	0.178	0.018	0.042	0.016	0.099
	q (BH)	0.284	0.054	0.119	0.054	0.189
NonGS-WTA	$\hat{\beta}_h$	6.2*	7.1*	10.5**	6.2	5.3
	p	0.087	0.063	0.011	0.293	0.517
	q (BH)	0.174	0.144	0.038	0.412	0.667
<i>Panel: Main draws</i>						
GS-ATP	$\hat{\beta}_h$	-0.2	0.1	0.1	0.7	0.8
	p	0.759	0.896	0.777	0.227	0.567
	q (BH)	0.822	0.907	0.829	0.343	0.721
GS-WTA	$\hat{\beta}_h$	-0.1	0.3	0.3	0.7	1.8
	p	0.813	0.618	0.652	0.331	0.247
	q (BH)	0.845	0.745	0.745	0.457	0.366
NonGS-ATP	$\hat{\beta}_h$	-1.4***	-0.9***	-0.3*	1.7***	5.2***
	p	0.000	0.000	0.076	0.000	0.000
	q (BH)	0.000	0.000	0.170	0.000	0.000
NonGS-WTA	$\hat{\beta}_h$	-1.1***	-0.7***	-0.4***	0.9***	2.7***
	p	0.000	0.000	0.008	0.000	0.000
	q (BH)	0.000	0.000	0.028	0.000	0.000
<i>Panel: Matches 250+</i>						
GS-ATP	$\hat{\beta}_h$	0.3	0.7	0.9	1.5	1.3
	p	0.799	0.462	0.340	0.169	0.650
	q (BH)	0.841	0.606	0.460	0.280	0.745
GS-WTA	$\hat{\beta}_h$	-0.5	0.8	0.6	1.9*	3.2
	p	0.588	0.288	0.428	0.055	0.205
	q (BH)	0.735	0.412	0.571	0.134	0.316
NonGS-ATP	$\hat{\beta}_h$	-2.6***	-1.7***	-0.6*	3.0***	8.9***
	p	0.000	0.000	0.055	0.000	0.000
	q (BH)	0.000	0.000	0.134	0.000	0.000
NonGS-WTA	$\hat{\beta}_h$	-2.1***	-1.4***	-0.8***	1.7***	5.6***
	p	0.000	0.000	0.006	0.000	0.000
	q (BH)	0.000	0.000	0.024	0.000	0.000

Notes. BH correction applied jointly across the family of 80 tests (4 primary outcomes $\times$ 4 samples $\times$ 5 horizons). Primary outcomes are ranking points, Elo, main draws entered, and matches at 250+ level events. $q < 0.05$ indicates significance after the joint correction. First-LL samples.

Table 33: News Audit of Ambiguous Lucky Loser Assignments

Tour	Year	Slam	LL entrant	Withdrawn player	Date
ATP	2014	Australian Open	Martin Klizan	David Goffin	not dated
ATP	2019	Wimbledon	Brayden Schnur	Borna Coric	not dated
ATP	2019	US Open	Kamil Majchrzak	Milos Raonic	2019-08-25
ATP	2021	Roland Garros	Francisco Cerundolo	Milos Raonic	2021-05-30
ATP	2023	Wimbledon	Fabian Marozsan	Nick Kyrgios	2023-07-02
ATP	2024	Wimbledon	Giovanni Mpetshi Perricard	Alejandro Davidovich Fokina	2024-06-30
WTA	2023	Australian Open	Leolia Jeanjean	Ajla Tomljanovic	2023-01-14
WTA	2023	Roland Garros	Camila Osorio	Paula Badosa (and prior withdrawals)	not dated
WTA	2023	US Open	Yanina Wickmayer	Bianca Andreescu / Paula Badosa	2023-08-26
WTA	2024	Wimbledon	Renata Zarazua	Ekaterina Alexandrova	2024-06-30
<i>Unresolved (no datable source found): 7 entries</i>					

Notes. Grand Slam first-time LL entries ranked first among qualifying losers, a subset of the 65 entries whose rank falls within the event’s slot count and whose assignment mechanism therefore depends on withdrawal timing not recorded in the data. Each row was checked against tennis-media reporting and tournament records for the date of the main-draw withdrawal that opened the slot. Every datable case was a post-qualifying withdrawal, which the ranking rule fills deterministically rather than by lottery. Entries marked “not dated” had corroborating reports of a late injury withdrawal without a precise date. The unresolved entries are predominantly pre-2015 events with thin contemporaneous coverage. The 42 ambiguous entries ranked second or lower have not been audited and are excluded from the verified subsample on the rank criterion alone.

Table 34: Romano-Wolf Step-Down p -Values Across Primary Outcomes

Sample		4w	8w	12w	26w	52w
<i>Panel: Ranking points</i>						
GS-ATP	$\hat{\beta}_h$	72.5***	72.1***	82.8***	72.6**	23.5
	q (BH)	0.028	0.028	0.009	0.126	0.780
	p (RW)	0.348	0.348	0.130	0.858	1.000
GS-WTA	$\hat{\beta}_h$	40.0	55.0**	27.8	7.5	26.5
	q (BH)	0.280	0.054	0.280	0.883	0.822
	p (RW)	0.985	0.558	0.986	1.000	1.000
NonGS-ATP	$\hat{\beta}_h$	20.6**	20.5**	14.1	44.6***	41.1
	q (BH)	0.119	0.126	0.280	0.008	0.280
	p (RW)	0.858	0.874	0.986	0.168	0.986
NonGS-WTA	$\hat{\beta}_h$	18.1	19.2*	23.2*	19.2	58.5
	q (BH)	0.209	0.174	0.135	0.409	0.264
	p (RW)	0.964	0.947	0.893	0.996	0.983
<i>Panel: Elo</i>						
GS-ATP	$\hat{\beta}_h$	-4.4	0.2	-4.8	4.7	-9.7
	q (BH)	0.745	0.986	0.745	0.812	0.745
	p (RW)	1.000	1.000	1.000	1.000	1.000
GS-WTA	$\hat{\beta}_h$	13.3*	15.0*	11.8	21.6*	36.8*
	q (BH)	0.174	0.174	0.306	0.182	0.134
	p (RW)	0.937	0.937	0.986	0.947	0.874
NonGS-ATP	$\hat{\beta}_h$	8.4	12.8**	11.4**	30.8**	33.6*
	q (BH)	0.284	0.054	0.119	0.054	0.189
	p (RW)	0.986	0.623	0.858	0.603	0.955
NonGS-WTA	$\hat{\beta}_h$	6.2*	7.1*	10.5**	6.2	5.3
	q (BH)	0.174	0.144	0.038	0.412	0.667
	p (RW)	0.947	0.905	0.488	0.996	1.000
<i>Panel: Main draws</i>						
GS-ATP	$\hat{\beta}_h$	-0.2	0.1	0.1	0.7	0.8
	q (BH)	0.822	0.907	0.829	0.343	0.721
	p (RW)	1.000	1.000	1.000	0.988	1.000
GS-WTA	$\hat{\beta}_h$	-0.1	0.3	0.3	0.7	1.8
	q (BH)	0.845	0.745	0.745	0.457	0.366
	p (RW)	1.000	1.000	1.000	0.997	0.992
NonGS-ATP	$\hat{\beta}_h$	-1.4***	-0.9***	-0.3*	1.7***	5.2***
	q (BH)	0.000	0.000	0.170	0.000	0.000
	p (RW)	0.000	0.001	0.939	0.000	0.000
NonGS-WTA	$\hat{\beta}_h$	-1.1***	-0.7***	-0.4***	0.9***	2.7***
	q (BH)	0.000	0.000	0.028	0.000	0.000
	p (RW)	0.000	0.001	0.415	0.004	0.001
<i>Panel: Matches 250+</i>						
GS-ATP	$\hat{\beta}_h$	0.3	0.7	0.9	1.5	1.3
	q (BH)	0.841	0.606	0.460	0.280	0.745
	p (RW)	1.000	1.000	0.997	0.985	1.000
GS-WTA	$\hat{\beta}_h$	-0.5	0.8	0.6	1.9*	3.2
	q (BH)	0.735	0.412	0.571	0.134	0.316
	p (RW)	1.000	0.996	1.000	0.874	0.986
NonGS-ATP	$\hat{\beta}_h$	-2.6***	-1.7***	-0.6*	3.0***	8.9***
	q (BH)	0.000	0.000	0.134	0.000	0.000
	p (RW)	0.000	0.001	0.892	0.000	0.000
NonGS-WTA	$\hat{\beta}_h$	-2.1***	-1.4***	-0.8***	1.7***	5.6***
	q (BH)	0.000	0.000	0.024	0.000	0.000
	p (RW)	0.000	0.001	0.348	0.018	0.001

Notes. Romano-Wolf step-down procedure (Romano and Wolf, 2005) over the same 80-test family as Table 32, computed with the `wildrwolf` implementation of Clarke et al. (2020) using a null-imposed wild cluster bootstrap with Webb weights, 9,999 replications, and clustering at the player level. Where Benjamini-Hochberg controls the false discovery rate and treats the tests as independent, Romano-Wolf controls the family-wise error rate while exploiting the joint dependence across horizons and outcomes, which is substantial here because outcomes at adjacent horizons are strongly autocorrelated. BH q -values are reproduced for comparison.

Table 35: Prize Money Effect of First-Time LL Entry (USD)

Sample	Immediate (USD)	Dynamic 26w (USD)	Total (USD)	Treated N
GS-ATP	\$93,770***	\$7,260	\$101,030	61
GS-WTA	\$86,667***	\$7,260	\$93,927	33
NonGS-ATP	\$9,000	\$18,810	\$27,810	523
NonGS-WTA	\$6,000	\$10,230	\$16,230	378

Notes. Immediate effect: difference in focal-event prize money between treated (LL recipients, earning based on main draw rounds won) and control (qualifying-round stipend). Dynamic 26w: estimated access gains (additional main draws entered by 26 weeks) evaluated at an average forward-looking first-round prize of \$11,000. Prize figures are approximate 2020–2024 averages; exact values vary by tournament tier and year. First-LL samples. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ on immediate effect t -test.

Table 36: Dose-Response: Non-Grand-Slam (First-LL)

	4w	8w	12w	26w	52w
<i>NGS-ATP</i>					
Ranking Pts Δ	9.3 (10.1)	1.1 (10.7)	-8.4 (10.7)	17.6 (15.1)	12.1 (32.2)
$\times$ Dose	22.6** (9.5)	37.0*** (8.5)	42.9*** (9.7)	52.4*** (18.0)	56.1* (29.3)
Main Draws	-1.2*** (0.2)	-0.8*** (0.2)	-0.3* (0.2)	1.4*** (0.3)	4.2*** (0.7)
$\times$ Dose	-0.3** (0.2)	-0.2 (0.1)	0.1 (0.1)	0.6** (0.3)	1.9*** (0.6)
Elo Δ	5.8 (6.4)	10.0* (5.7)	8.1 (5.9)	32.1** (13.5)	35.1* (21.3)
$\times$ Dose	4.6** (2.3)	5.0* (2.7)	6.2* (3.3)	-2.4 (4.7)	-2.4 (5.5)
<i>NGS-WTA</i>					
Ranking Pts Δ	3.9 (12.5)	0.9 (12.4)	5.4 (13.9)	7.9 (20.3)	45.8 (44.5)
$\times$ Dose	30.0*** (7.9)	37.2*** (8.5)	37.4*** (9.6)	25.0 (16.3)	33.1 (36.5)
Main Draws	-1.0*** (0.2)	-0.6*** (0.1)	-0.3** (0.1)	0.7*** (0.2)	2.2*** (0.6)
$\times$ Dose	-0.1 (0.1)	-0.1 (0.1)	-0.0 (0.1)	0.4** (0.2)	1.3** (0.6)
Elo Δ	3.7 (3.8)	5.3 (4.0)	9.0** (4.4)	6.7 (6.5)	10.8 (9.2)
$\times$ Dose	4.7* (2.4)	3.4 (2.6)	3.0 (3.4)	-1.0 (5.4)	-12.6 (9.5)

Notes. Treatment interacted with matches won at the LL event. First-LL non-GS samples with CF. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.